

DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO**TEST BOOKLET SERIES****Test Code:
2209****TEST BOOKLET
GENERAL STUDIES
TEST – 9****A****Time Allowed : Two Hours****Maximum Marks : 200****INSTRUCTIONS**

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET *DOES NOT* HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC. IF SO, GET IT REPLACED BY A COMPLETE TEST BOOKLET.
2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE **A, B, C** OR **D** AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.
3. You have to enter your Roll Number on the Test Booklet in the Box provided alongside this line. *DO NOT* write *anything else* on the Test Booklet.
4. This Test Booklet contains 100 items (questions). Each question is printed both in **Hindi** and **English**. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose **ONLY ONE** response for each item.
5. You have to mark all your responses **ONLY** on the separate Answer Sheet provided. See directions in the Answer sheet.
6. All items carry equal marks.
7. Before you proceed to mark in the Answer Sheet the response to various items in the Test Booklet, you have to fill in some particulars in the Answer Sheet as per instructions sent to you with your admission certificate.
8. After you have completed filling in all your responses on the Answer Sheet and examination has concluded, you should hand over to the Invigilator *only the Answer Sheet*. You are permitted to take away with you the Test Booklet.
9. Sheets for rough work are appended in the Test Booklet at the end.
10. **Penalty for wrong answers:**
THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.
 - (i) There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, **One-Third** of the marks assigned to that question will be deducted as penalty.
 - (ii) If a candidate gives more than one answer, it will be treated as a **wrong answer** even if one of the given answers happens to be correct and there will be same penalty as above to that question.
 - (iii) If a question is left blank, i.e., no answer is given by the candidate, there will be no **penalty** for that question.

DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO*Note:* Hindi version of the instructions is printed on the front cover of this Booklet.



1. Which of the following are the outcomes of the United Nations Conference On Environment And Development (UNCED), 1992?

1. Rio Declaration on Environment and Development
2. Agenda 21
3. Forest Principles
4. Convention on Biological Diversity
5. United Nations Framework Convention on Climate Change (UNFCCC)

Select the correct answer using the code given below:

- (a) 1, 3 and 5 only (b) 1, 2 and 3 only
(c) 1, 2, 4 and 5 only (d) 1, 2, 3, 4 and 5

2. With reference to Canine Distemper Virus (CDV), consider the following statements:

1. It is responsible for recent tiger deaths in the Kanha National Park.
2. This virus spreads through the dogs to wild animals.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

3. With reference to Loan Guarantee Scheme for Covid-Affected Sectors (LGSCAS) consider the following statements:

1. The main objective is to partially mitigate credit risk and facilitate bank credit at lower interest rates.
2. Banks loans for building healthcare infrastructure in non-metro cities and smaller towns at an interest rate of 7.95%

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

4. Consider the following statements:

1. A keystone species will never have a symbiotic relationship with other species.
2. Estuary is an example of a keystone ecosystem.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

5. What is Greenwashing?

- (a) It is a practice of misleading claim about the environmental benefits of a product.
- (b) It is a practice of coal washing to reduce fumes generated in thermal plants.
- (c) It is a method for generation of fuel oil from solid green waste.
- (d) It is an biodegradable polythene developed by Indian scientists.

6. With reference to Resolution 2593 of UNSC, consider the following statements:

1. It reiterates the importance of combating terrorism in Afghanistan, including those individuals and entities designated according to resolution 1267 (1999).
2. China abstained from voting for the resolution as it believes that the current chaos results from Western countries' "disorderly withdrawal".
3. India-led United Nations Security Council (UNSC) adopted Resolution 2593 on the Taliban.

Which of the statements given above is/are correct?

- (a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2, and 3

7. Which of the following conventions is **not** affiliated with any of the United Nations systems of Multilateral Environmental Agreements?

- (a) Convention on Migratory Species (Bonn Convention)
- (b) Convention of Wetlands (Ramsar Convention)
- (c) Vienna Convention for protection of ozone layer
- (d) Rotterdam convention



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8. With reference to the World Social Protection Report, consider the following statements:
1. It is published by the International Labour Organization.
 2. Limited Unemployment Protection and Barriers to Accessing Healthcare are some of the report's highlights.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
9. Consider the following statements about the Invasive Alien species:
1. Invasive alien species are able to grow rapidly as compared to other species due to relatively smaller cellular DNA content.
 2. Naturalized species have the potential to become an Invasive alien species.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
10. With reference to the e-ILP system, consider the following statements:
1. It is issued by the concerned State Government and can solely be issued for travel purposes.
 2. Foreigners need a Protected Area Permit (PAP) to visit places different from ILPs required by domestic tourists.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
11. The term 'Biomining', refers to:
- (a) Use of microorganism to extract metals
 - (b) Harvesting stem cells
 - (c) Blood transfusion
 - (d) None of the above
12. Consider the following statements regarding the United Nations-Food and Agriculture Organisation (FAO):
1. The day of establishment of FAO is celebrated as World Food Day.
 2. FAO launched the Desert Locust Programme in 1952 on the request of India.
 3. It has launched the UN Decade of Family Farming from 2019-2028.
- Which of the statements given above is/are correct?
- (a) 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3
13. Regarding the Convention on the Conservation of Migratory Species of Wild Animals, consider the following statements:
1. It is an environmental treaty of the United Nations.
 2. Migratory species threatened with extinction are listed in Appendix I of the Convention.
 3. All agreements under the convention are legally binding on all the parties.
- Which of the statements given above are correct?
- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
14. Consider the following statements:
1. The World Meteorological Organisation is a specialised agency of the United Nations.
 2. Sendai Framework 2015 was adopted at the Third United Nations World Conference on Disaster Risk Reduction in 2015 in Sendai, Miyagi, Japan.
- Which of the statements given above is/are correct?
- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2





15. Which of the following species of vultures is **not** Critically Endangered?
- Long-billed Vulture
 - Slender-billed Vulture
 - Oriental White-backed Vulture
 - Egyptian Vulture
16. Which one of the following organisations released the State of the World's Trees Report?
- World Wildlife Fund (WWF)
 - United Nation Environment Programme
 - International Union for Conservation of Nature
 - Botanic Gardens Conservation International
17. Consider the following statements:
- A larger area usually has more species as compared to a smaller area of similar habitat.
 - Organisms of a trophic level have similar ways of life.
- Which of the statements given above is/are correct?
- 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2
18. With reference to the International Fund for Agricultural Development (IFAD), consider the following statements:
- It is an international financial institution and specialized United Nations agency working in poverty eradication in the rural areas of developing countries.
 - It brings out the Rural Development Report every year.
 - It was established in 1967, as an outcome of the World Food Conference of 1964.
 - It has 177 member countries and India is not a member country.
- Which of the statements given above is/are **not** correct?
- 1 and 2 only
 - 2 and 3 only
 - 3 and 4 only
 - 1, 2 and 4 only
19. With reference to the Petroleum Coke, consider the following statements:
- It emits more greenhouse gases than coal.
 - Petroleum coke has higher energy content than coal.
- Which of the statements given above is/are correct?
- 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2
20. Which of the following describes best the planting of trees on the sides of roads, canals and railways, along with planting on wastelands?
- Recreational forestry
 - Community forestry
 - Extension forestry
 - None of the above
21. With reference to the Account Aggregator (AA) network, consider the following statements:
- AA is a framework that facilitates sharing of financial information in a real-time and data-blind manner between regulated entities.
 - It permits users to control who gets access to their data, track and log its movement and reduce the potential risk of leakage in transit.
 - AA reduces the fraud associated with biological data by introducing secure digital signatures and end-to-end encryption for data sharing.
- Which of the statements given above is/are correct?
- 1 and 2 only
 - 3 only
 - 2 and 3 only
 - 1, 2 and 3



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22. With reference to the Common Services Centre, consider the following statements:

1. The CSC scheme's PPP (Public-Private Partnership) model is a 3-tier structure.
2. It is a strategic cornerstone of the National e-Governance Plan (NeGP), as part of its commitment to the National Common Minimum Programme to introduce e-governance on a massive scale.
3. CSC 2.0 was established in 2015, and at least one CSC is envisaged in each 2.5 lakh Gram Panchayats.

Which of the statements given above is/are correct?

- (a) 1 only (b) 3 only
(c) 2 and 3 only (d) 1, 2, and 3

23. With reference to the draft notification of the Western Ghats Ecologically Sensitive Area (ESA), consider the following statements:

1. Gadgil Committee recommended 37 percent of total Western Ghats to be declared as eco-sensitive.
2. The Eco-Sensitive Zone is the fragile areas located within the Protected Areas, National Parks and Wildlife Sanctuaries.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

24. With reference to the textile sector's Production Linked Incentive (PLI) scheme, consider the following statements:

1. It aims to promote the production of high-value Man-Made Fibre fabrics and technical textiles.
2. PLI for Textiles will herald a new age in textiles manufacturing.
3. It will lead to more than 7.5 lakh jobs in this sector and several lakhs more for supporting activities.
4. This scheme will positively impact states like Gujarat, UP, Maharashtra, Odisha etc.

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 3 and 4 only (d) 1, 2, 3 and 4

25. Which of the following major missions are covered under India's National Action Plan on Climate Change (NAPCC)?

1. National Water Mission
2. Green India Mission
3. National Mission for Sustaining the Himalayan Ecosystem
4. National Mission for Clean Ganga
5. National Mission for Enhanced Energy Efficiency

Select the correct answer using the code given below:

- (a) 2, 3 and 5 only (b) 1, 2, 3 and 5 only
(c) 1, 3, 4 and 5 only (d) 1, 2, 3, 4 and 5

26. With reference to Dugong, consider the following statements:

1. It is the only herbivorous mammal that lives exclusively in the sea.
2. In India, they are found in the waters of Andaman and Nicobar Islands only.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

27. Atmanirbhar Bharat Corner is an initiative of which the following ministry?

- (a) Ministry of External Affairs
(b) Ministry of Commerce and Industry
(c) Ministry of Tribal Affairs
(d) Ministry of Finance

28. With reference to report on 'Heavy Metals Contaminating Indian Rivers' by Central Water Commission (CWC), which of the following has been found to be the most common contaminant?

- (a) Lead (b) Iron
(c) Arsenic (d) Cadmium



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29. Which of the following agencies/institutions publishes the Brown to Green Report?
- UNCCD
 - UNFCCC
 - Climate Transparency
 - Greenpeace
30. Consider the following statements:
- Under the Transport and Marketing Assistance scheme, the government provides financial assistance for the transport and marketing for agricultural produce.
 - Agricultural and Processed Food Products Export Development Authority has been mandated with the responsibility of export promotion and development of the scheduled products.
- Which of the statements given above is/are correct?
- 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2
31. Consider the following statements about National Clean Air Programme (NCAP):
- The NCAP focusses on reducing particulate matter (PM) 2.5 only.
 - The plan will be implemented in cities which featured in 20 most polluted cities of the world.
- Which of the statements given above is/are correct?
- 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2
32. Consider the following statements:
- Rainforest ecosystems can be found on all the continents except Antarctica.
 - Deciduous forests are Earth's oldest surviving ecosystems.
- Which of the statements given above is/are correct?
- 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2
33. With reference to Strategic Clean Energy Partnership (SCEP), consider the following statements:
- Emerging fuels are added as the fifth pillar of cooperation in SCEP.
 - The SCEP was launched under the US - India Climate and Clean Energy Agenda 2030.
- Which of the statements given above is/are correct?
- 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2
34. Consider the following statements:
- The first National Wildlife Action Plan (NWAP) was adopted in 1982.
 - The National Afforestation Programme involves plantation only in degraded forests of the country.
- Which of the statements given above is/are correct?
- 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2
35. Recently, which of the following have been included in Appendix I of the UN Convention on Migratory Species on India's proposal?
- The Great Indian Bustard
 - Asian Elephant
 - Bengal Florican
- Select the correct answer using the code given below:
- 1 and 3 only
 - 2 and 3 only
 - 1 and 2 only
 - 1, 2 and 3



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36. Regarding Swachh Survekshan-Grameen 2021, consider the following statements:

1. It is an initiative of Ministry of Jal Shakti.
2. It aims to assesses the state of hygiene, cleanliness and sanitation in rural India.
3. Swachh Bharat Mission (Grameen) Phase II emphasises the sustainability of achievements under phase I and facilitates Solid/Liquid & plastic Waste Management (SLWM).

Which of the following statements is/are correct?

- (a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3

37. With reference to the Convention on Biological Diversity (CBD), consider the following statements:

1. It is a legally binding multilateral environmental agreement.
2. CBD is empowered to facilitate access to the genetic resources of a country by other parties subject to national legislation and on mutually agreed terms.

Which of the statements given above is/are correct?

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

38. With reference to Colour Coded Weather Warning by IMD, consider the following statements:

1. 'Yellow' weather warning indicates severely bad weather spanning across several days.
2. The 'Orange' alert is issued as a warning of extremely bad weather with the potential of disruption in rail and road communication.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

39. With reference to Trade Credit Insurance, consider the following statements:

1. It contributes to the economic growth of a country
2. It applies to all insurers transacting general insurance business, registered under the Insurance Act, 1938
3. It enable general insurance for the Micro, Small and Medium Enterprises (MSMEs)
4. It protects businesses against the risk of non-payment for goods and services

Which of the statements given above are correct?

- (a) 1 and 3 only
(b) 2 and 4 only
(c) 1, 2 and 3 only
(d) 1, 2, 3 and 4

40. Consider the following statements regarding the Satkosia Tiger Reserve:

1. Satkosia Tiger Reserve is located on the bank of Subarnarekha river.
2. It lies between a transitional zone extending between the Chhota Nagpur Plateau and the Deccan Plateau.

Which of the statements given above is/are correct?

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

41. Consider the following statements:

1. The coral triangle is located in the Caribbean sea.
2. Decrease in salinity of ocean water harms the coral density of this area.

Which of the statements given above is/are correct?

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2



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42. With reference to the Groundswell report, consider the following statements:

1. A Groundswell report released by the World Bank indicates the climate change that could force people to move within their countries by 2050.
2. First Groundswell Report was published in 2018, focusing on sub-Saharan Africa, South Asia, and Latin America.
3. The recommendation in this report includes Reduced Emission and Inclusive and Resilient Development Pathways.
4. Cancun Adaptation Framework (2010) by United Nations Framework Convention on Climate Change is one of the Global Efforts to Address Climate Change.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 and 4 only
(c) 1, 2 and 3 only (d) 1, 2, 3 and 4

43. Regarding the UNESCO's Global Geopark, consider the following statements:

1. These are sites and landscapes of international geological significance which are managed with a holistic concept of protection, education and sustainable development.
2. Lonar lake is the only UNESCO Global Geopark in India.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

44. Consider the following statements:

1. Coliform bacteria act as an indicator of water pollution.
2. Biochemical Oxygen Demand is the amount of dissolved oxygen used by microorganisms in the biological process in water.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

45. Yellow Rust disease appears as yellow stripes of powder or dust on leaves and leaf sheaths of the wheat crop. Which of the following environmental conditions can provide favourable conditions for this disease to infest the crops?

1. Rain
2. Dew conditions
3. Foggy weather
4. Extreme Heating conditions

Select the correct answer using the code given below:

- (a) 1, 2 and 3 only
(b) 2, 3 and 4 only
(c) 1, 3 and 4 only
(d) 1, 2, 3 and 4

46. With reference to Deepfakes, consider the following statements:

1. Deepfake is a synthetic media created with the help of Artificial Intelligence.
2. The existing legal framework of India does not criminalize deep fakes.

Which of the statements given above is/are correct?

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

47. Consider the following statements about Biomes:

1. Taiga biome is devoid of trees and the flora here includes lichens and mosses.
2. Low rainfall and wild fires disrupt the maintenance of grasslands.

Which of the statements given above is/are correct?

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2



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48. With reference to 'Pradhan Mantri Poshan Shakti Nirman', consider the following statements:

1. The scheme has replaced the 'National Programme Mid-Day Meal Scheme'.
2. It will provide additional supplements to aspirational districts and children prone to anaemia.
3. The Balvatika children are also covered in this scheme to protect children from malnutrition.
4. Vocal for Local will also be encouraged and promoted with the scheme's implementation.

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1, 2 and 3 only (d) 1, 2, 3 and 4

49. To transform Indian Agriculture for global environment benefits and the conservation of critical biodiversity and forest landscapes, Green Ag program has been launched by the Indian government. Which of the following international organisations has assisted in the launch of this program?

- (a) Green Climate Fund
(b) Global Environment Facility
(c) United Nations Environment Programme
(d) Intergovernmental Panel on Climate Change

50. With reference to Export Credit Guarantee Corporation (ECGC), consider the following statements:

1. The union cabinet approved capital infusion to the ECGC for three years.
2. It also approved the continuation of the National Export Insurance Account (NEIA) with a 1,650 crores grant-in-aid.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

51. Regarding the climate vulnerability maps developed for Himalayan states, consider the following statements:

1. The vulnerability index is the highest for the union territory of Jammu and Kashmir.

2. Low per capita income and large number of marginal farmers are the probable reasons behind high vulnerability to climate change in the Himalayan region.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

52. Consider the following statements regarding the INSPIRESAT-1 Cubesat:

1. This satellite will be launched by the Indian Space Research Organisation (ISRO).
2. The INSPIRE program aims to provide a constellation of Earth and space weather-observing satellites.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

53. With reference to The Future of Earth-2020 Report, consider the following statements:

1. It has been jointly released by the South Asia Future Earth Regional Office, and the Indian Institute of Science.
2. The report has been prepared with the aim of reducing carbon footprint and halting global warming below 2 degree Celsius by 2050.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

54. Which of the following is/are the features of the Drone Rules 2021?

1. The coverage of drones has been decreased from 500 kg to 300 kg.
2. The maximum penalty has been reduced to Rs.1 lakh.
3. The yellow zone around the airport boundary, has been decreased from 45 km to 12 km.

Select the correct answer using the code given below:

- (a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3



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55. Consider the following statements about Pygmy Hog:

1. Pygmy Hog can be found in the Manas National Park, Assam.
2. It has not been included in the Species Recovery Programme for critically endangered species.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

56. Consider the following statements with reference to National Disaster Management Authority (NDMA):

1. NDMA is a constitutional body for disaster management in India.
2. It is headed by Prime Minister of India.
3. It is implementing the Central Sector Scheme - Aapda Mitras Programme.

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

57. Consider the following statements regarding the Scheduled Tribes and other traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006:

1. The Union Ministry of Environment, Forests and Climate change is the nodal agency for the implementation of this act
2. According to this act, the maximum limit of the rights to hold or live on forest land is 6 hectares
3. The National Parks and Sanctuaries have been included along with Reserve Forest, Protected Forests for the recognition of Rights

Which of the statements given above is/are correct?

- (a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 3 only

58. With reference to the Akash Prime Missile, consider the following statements:

1. It has been built for performance under low-temperature and high altitude environments.
2. Akash Missile is a part of the Integrated Guided Missile Development Programme.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

59. With reference to 'Cartagena Protocol', consider the following statements:

1. It is a supplementary agreement to the United Nations Convention on Biological Diversity (CBD).
2. It establishes legally binding procedures for regulating the import and export of Living Modified Organisms (LMOs) from one country to another.
3. The protocol has provisions for an Advance Informed Agreement (AIA) procedure.
4. India is a party to the Cartagena Protocol.

Which of the statements given above are correct?

- (a) 1, 2 and 3 only (b) 1, 2 and 4 only
(c) 2 and 3 only (d) 1, 2, 3 and 4

60. Which of the following became the first traditional brew in the northeast to receive the Geographical Indication (GI) tag?

- (a) Judima (b) Chokuwa
(c) Pokkali (d) Chinnor

61. Which of the following terms correctly describes-entire sequence of communities in a given area, succeeding each other?

- (a) Pioneer Species
(b) Climax Species
(c) Xerarch
(d) Sere



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62. In which of the following straits did the third edition of Samudra Shakti, a bilateral maritime between India and Indonesia, occur?

- (a) Malacca strait
- (b) Lombok strait
- (c) Sunda strait
- (d) Karimata strait

63. Which of the following Indian wetland(s) forms the part of the Montreux record?

- 1. Keoladeo National Park
- 2. Chilika Lake
- 3. Nalsarovar Bird Sanctuary

Select the correct answer using the code given below:

- (a) 1 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

64. Which of the following small states topped in the 3rd State Food Safety Index?

- (a) Meghalaya
- (b) Goa
- (c) Sikkim
- (d) Manipur

65. With reference to 'The Global Tiger Forum (GTF)', consider the following statements:

- 1. It is the only inter-governmental international body campaigning to save the Tiger worldwide.
- 2. A Chairperson, usually a Minister from one of the Tiger Range countries heads GTF for a fixed tenure of 5 Years.
- 3. The Secretariat of GTF is located in New Delhi, India.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 1 and 3 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

66. Consider the following statements with reference to Global Innovation Index (GII) 2021:

- 1. GII was launched by the World Intellectual Property Organisation (WIPO).
- 2. In this Index, assessment of 132 economies was done with 81 different indicators.

3. According to Index, world's most innovative economy in 2021 is Switzerland, followed by the United States of America.

4. India has climbed up to 46th rank.

Which of the statements given above is/are **not** correct?

- (a) 1 only
- (b) 3 only
- (c) 2 and 3 only
- (d) 1 and 4 only

67. Consider the following statements about forest fires:

- 1. Eucalyptus trees are known to inhibit forest fires due to their moist branches and waxy leaves.
- 2. Surface fire type of forest fire is most common in India.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

68. Which of the following are recognised as "Three Sisters" by the World Trade Organization's (WTO) Sanitary and Phytosanitary Measures (SPS) Agreement?

- 1. International Plant Protection Convention (IPPC)
- 2. Codex Alimentarius Commission
- 3. Pacific Plant Protection Organization
- 4. World Organization for Animal Health (OIE)

Select the correct answer using the code given below:

- (a) 1, 2 and 3 only
- (b) 2, 3 and 4 only
- (c) 1, 2 and 4 only
- (d) 1, 3 and 4 only

69. With reference to Global Agricultural Productivity Report 2021, consider the following statements:

- 1. Globally, Total Factor Productivity (TFP) grew by an average of 1.36 per cent annually.
- 2. Low-income countries have a negative TFP growth rate.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2



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70. With reference to the weathering, consider the following statements:

1. Weathering done by Salts is known as the Haloclasty and it is a type of mechanical weathering.
2. Chemical weathering by water transforms anhydrite into the gypsum.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

71. Which of the following are unattached organisms which live at the air water interface?

- (a) Periphyton (b) Neuston
(c) Nekton (d) Benthos

72. With reference to Rail Kaushal Vikas Yojna, consider the following statements:

1. The scheme was launched by the Ministry of Railways and the Ministry of Skill Development and Entrepreneurship.
2. It is a skill development program.
3. There will be no reservations to candidates under this scheme
4. Candidates trained under this scheme can claim direct employment in Railways.

Which of the statements given above is/are **not** correct?

- (a) 2 only (b) 4 only
(c) 1, 2 and 3 only (d) 3 and 4 only

73. The 'World Energy Outlook' report is published by:

- (a) International Renewable Energy Agency
(b) International Energy Agency
(c) Organisation for Economic Co-operation and Development
(d) World Economic Forum

74. With reference to the Skinks, consider the following statements:

1. In India, skinks are found in the Himalayas as well as in the coastal region.

2. 62 species of skinks are found in India and about 57% of them (33 species) are endemic.
3. It is the largest family of lizards, having long bodies, relatively small or no legs.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

75. Consider the following statements:

1. This National Park/Wildlife Sanctuary has Nilgiri tahr as the keystone species.
2. It is located in Western Ghats and it is also the part of a UNESCO World Heritage Site.

Which of the following National Park/Wildlife Sanctuary is being referred in the statements given above?

- (a) Bannerghatta National Park – Karnataka
(b) Pobitora Wildlife Sanctuary – Assam
(c) Silent Valley National Park – Kerala
(d) Mukurthi National Park – Tamil Nadu

76. With reference to Shooonya Campaign, consider the following statements:

1. It is an initiative to promote digital payments across the country.
2. This campaign has been launched by NITI Aayog.

Which of the statements given above is/are correct?

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

77. Which of the following Tiger Reserve of India have the 'Black Tiger', which was recently in the news?

- (a) Sariska Tiger Reserve
(b) Pilibhit Tiger Reserve
(c) Satkosia Tiger Reserve
(d) Similipal Tiger Reserve



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78. Identify the conservation area from the facts given below:

1. It is a man-made wetland.
2. It is located on the confluence of River Banganga and Gambhir.
3. It has been recognized as a Ramsar site.

The statements given above describe?

- (a) Ropar Wetland
- (b) Bhoj Wetland
- (c) Nangai Wildlife Sanctuary
- (d) Keoladeo National Park

79. Where did the 6th Edition of Shanghai Cooperation Organisation (SCO) Exercise Peaceful Mission was held?

- (a) Russia
- (b) Kazakhstan
- (c) China
- (d) India

80. With reference to Mangroves, consider the following statements:

1. Mangroves can survive in high salinity, high temperature and muddy anaerobic soil.
2. Mangroves for the Future is joint programme of IUCN and UNDP.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

81. With reference to ECO-MARK, consider the following statements:

1. The scheme is being operated on a national basis and provides certification and labeling for household and other consumer products.
2. ECO-MARK is administered by the Bureau of Indian Standards.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

82. With reference to MeghEA, consider the following statements:

1. The initiative envisions making Meghalaya a high-income state by 2030.
2. It is spread along five pillars, i.e. Governance, Human Resources, Entrepreneurship, Primary Sector, and Infrastructure.

Which of the statements given above is/are **not** correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

83. Which of the following statements is correct about the term 'Arakunomics'?

- (a) an economic system in which transactions between private parties are absent of any form of economic interventionism such as regulation.
- (b) a new integrated economic model that ensures profits for farmers, quality for consumers through regenerative agriculture.
- (c) a theory that says the government should increase demand to boost growth
- (d) None of the above.

84. 'PLAN BEE', of the Indian Railways is related to:

- (a) Reducing delays of passenger trains
- (b) Safety of wild elephants
- (c) Increasing the energy efficiency of railway stations
- (d) Providing formal education and skilling to destitute children found on railway premises

85. With reference to the Scheme for "Promotion of Medical Device Parks", consider the following statements:

1. It is an initiative to support medical devices.
2. It aims to reduce manufacturing costs, optimise resources built on economies of scale and provide easy access to standard testing and infrastructure facilities.
3. The total financial outlay of the scheme is Rs. 400 crores, and the scheme's tenure is being implemented from 2020-2030.

Which of the statements given above is/are **not** correct?

- (a) 2 only
- (b) 3 only
- (c) 1 and 3 only
- (d) 2 and 3 only



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86. With reference to the Marine Protected Areas (MPAs), consider the following statements:

1. These are the zones where no human activities are permitted.
2. The International Union for Conservation of Nature engages in the policy making to promote the expansion of Marine Protected Areas (MPAs).

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

87. With reference to the Rajaji Tiger Reserve, consider the following statements:

1. It is located in the Garhwal Himalayas.
2. Sal (*Shorea Robusta*) is the characteristic tree species in this reserve.
3. Ganga and Song rivers flow through this park.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 2 only

88. With reference to CAMPA (Compensatory Afforestation Fund Management and Planning Authority), consider the following statements:

1. It was created by the Ministry of Environment, Forests and Climate Change in 2001 to compensate for deforestation.
2. The funds under CAMPA are to be utilised for Afforestation only.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

89. Consider the following statements about Ecological Pyramids:

1. Pyramid of energy is always upright.
2. Pyramid of biomass in sea is generally inverted.
3. Limitation of ecological pyramid is that it does not take into account the same species belonging to two or more trophic levels.

Which of the statements given above is/are correct?

- (a) 1, 2 and 3
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1 only

90. Consider the following statements:

1. Tarballs are dark-coloured, sticky balls of oil that form when crude oil.
2. Tarballs are formed by weathering of crude oil in marine environments.
3. Tarballs are coin-sized and weigh up to 2 kg.

Which of the statements given above is/are **not** correct about Tarballs?

- (a) 1 only (b) 3 only
(c) 2 and 3 only (d) 1 and 3 only

91. With reference to changes in the ecosystem caused by Eutrophication, consider the following statements:

1. Increased production of Algae and aquatic plant
2. Depletion of fish species
3. Deterioration of water quality

Which of the statements given above is/are correct?

- (a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3

92. With reference to the National Board for Wildlife (NBWL), consider the following statements:

1. It is constituted under the Wildlife Protection Act, 1972.
2. The NBWL is chaired by the Prime Minister.
3. NBWL recommendations are binding on the Central and State governments.

Which of the statements given above are correct?

- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3



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93. With reference to Web 3.0, consider the following statements:

1. The founder of Web 3.0 is Dr Aeron Buchanan.
2. The users of Web 3.0 are in control of their own data.
3. Web 3.0 provides faster and more relevant results because of its human-like distinguishing capacity.

Which of the statements given above is/are correct?

- (a) 2 only
- (b) 1 and 3 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

94. Consider the following statements about Agasthyamala Biosphere Reserve:

1. The reserve is home to Kanikaran tribe.
2. Annamalai Tiger Reserve is located inside Agasthyamala Biosphere Reserve.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

95. With reference to Rhizofiltration, consider the following statements:

1. It is a soil remediation technique by using the rhizobium bacteria.
2. It involves the uptake of contaminants by plant roots.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

96. Consider the following statements:

1. The First Movers Coalition is a public-private partnership between the US State Department and the World Economic Forum.

2. The First Movers Coalition aims to unlock the untapped potential of emerging technologies needed to decarbonise the world by 2050.
3. The first phase of the First Movers Coalition will focus on "hard-to-abate" sectors.

Which of the statements given above is/are correct?

- (a) 2 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

97. In which of the following sites can the endangered Red Panda can be spotted?

1. Namdhapha National Park
2. Neora Valley National Park
3. Khangchendzonga Biosphere Reserve

Select the correct answer using the code given below:

- (a) 1 and 3 only
- (b) 1, 2 and 3
- (c) 2 and 3 only
- (d) 1 and 2 only

98. Photochemical smog has become a regular phenomena in urban areas. Which of the following plant/plants can help in reducing the photochemical smog?

1. Pinus
2. Juniperus
3. Peepal

Select the correct answer using the code given below:

- (a) 2 and 3 only
- (b) 2 only
- (c) 1 and 2 only
- (d) 1 only



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99. With reference to Ramon Magsaysay awards, consider the following statements:

1. The Ramon Magsaysay Award is considered Asia's premier prize and highest honour.
2. The award is internationally-recognized as the Nobel Prize counterpart of Asia.
3. Recognises and honours individuals and organisations in Asia who have helped others generously without aiming for public recognition.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

100. Consider the following statements:

1. India has the largest number of Ramsar sites in South Asia.
2. Asan Conservation Centre, is a Ramsar site on Ganga river.
3. Kabartal Wetland is an important stopover site along the Central Asian Flyway.

Which of the statements given above are correct?

- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

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Test Code:
2209

(English Version)

ANSWER & EXPLANATION
GENERAL STUDIES
TEST – 09
A
ANSWERS

Q. No.	Answers	Q. No.	Answers	Q. No.	Answers	Q. No.	Answers
1	(d)	26	(a)	51	(b)	76	(b)
2	(b)	27	(a)	52	(c)	77	(d)
3	(c)	28	(b)	53	(c)	78	(d)
4	(b)	29	(c)	54	(c)	79	(a)
5	(a)	30	(c)	55	(c)	80	(c)
6	(d)	31	(d)	56	(b)	81	(c)
7	(b)	32	(a)	57	(d)	82	(b)
8	(c)	33	(c)	58	(c)	83	(b)
9	(c)	34	(b)	59	(d)	84	(b)
10	(c)	35	(d)	60	(a)	85	(b)
11	(a)	36	(d)	61	(d)	86	(b)
12	(d)	37	(c)	62	(c)	87	(b)
13	(a)	38	(c)	63	(a)	88	(d)
14	(c)	39	(d)	64	(b)	89	(a)
15	(d)	40	(b)	65	(b)	90	(b)
16	(d)	41	(b)	66	(b)	91	(d)
17	(c)	42	(d)	67	(b)	92	(a)
18	(c)	43	(a)	68	(c)	93	(d)
19	(c)	44	(c)	69	(c)	94	(a)
20	(c)	45	(a)	70	(c)	95	(b)
21	(d)	46	(c)	71	(b)	96	(d)
22	(d)	47	(d)	72	(b)	97	(b)
23	(d)	48	(d)	73	(b)	98	(c)
24	(d)	49	(b)	74	(d)	99	(d)
25	(b)	50	(b)	75	(d)	100	(c)

**1. (d)****Explanation:**

- The United Nations Conference On Environment and Development (UNCED) is also known as the **Rio Summit**, Rio Conference, **Earth Summit** held in Rio de Janeiro, Brazil in June 1992.
- The Earth Summit resulted in the following documents:
 1. Rio Declaration on Environment and Development
 2. Agenda 21
 3. Forest Principles
- UNCED also resulted in two important **legally binding** agreements:
 1. Convention on Biological Diversity
 2. Framework Convention on Climate Change (UNFCCC).

Hence, statements 1, 2, 3, 4 and 5 are correct.

2. (b)**Explanation:**

- The Indian Council for Medical Research (ICMR) has confirmed that the Canine Distemper Virus (CDV) was responsible for **lion deaths** in the **Gir forest** of Gujarat and recommended that the remaining lions be vaccinated to prevent further outbreaks. **Hence, statement 1 is NOT correct.**
- This goes against recommendations by wildlife biologists that wild animals shouldn't be vaccinated as it may compromise their immunity against future infections.
- Laboratory tests have confirmed that four of the 23 lions had been infected by the Canine Distemper Virus (CDV) while 10 others had been infected by Babesia protozoa, an infection spread by tick parasite.
- Apart from vaccination, ICMR has also recommended that the animals should be placed in 2-3 different sanctuaries.
- The devastating link between the CDV and potential epidemics has been known since 1994, when within the Serengeti-Mara ecosystem of East Africa (Tanzania) nearly a third of the lion population died or disappeared.
- The same pattern is being repeated in Gujarat today, with the Forest Department confirming the spread of the CDV from dogs to the lions.

Canine Distemper Virus:

- Canine distemper virus is known mainly to cause a severe infection in dogs respiratory, gastrointestinal, respiratory and central nervous systems, as well as the conjunctival membranes of the eye.
- CDV may also affect wild carnivores such as wolves, foxes, raccoons, red pandas, ferrets, hyenas, tigers, and lions.
- The prevalence of this virus and its diversity in wildlife of India is not adequately studied.
- **A lion does not eat the entire prey at one go. In between, the dogs consume the kill and infect it with the CDV. Once the lion returns to finish it off, it gets the deadly disease. Hence, statement 2 is correct.**
- The CDV is more dangerous for lions than tigers. This is because lions move together in large numbers, making them more vulnerable to the virus as compared to tigers that are more isolated and territorial animals.

3. (c)**Explanation:****Loan Guarantee Scheme for Covid-Affected Sectors (LGSCAS):**

- The Finance Minister urged all stakeholders to conclude the **Rs 50,000 crore Loan Guarantee Scheme for Covid-Affected Sectors (LGSCAS)** in non-metros in time to deal with any future third wave of COVID-19.
- Approved in June 2021, it aims at boosting medical infrastructure in the country, explicitly targeting under-served areas. Under this, the banks would provide loans for building healthcare infrastructure in non-metro cities and smaller towns at a maximum interest rate of 7.95%. **Hence, statement 2 is correct.**
- LGSCAS guarantees 50% for brownfield projects and 75% for greenfield projects for loans sanctioned up to Rs.100 crore, set up at urban or rural locations other than 8 Metropolitan Tier 1 cities (Class X cities). The guarantee cover for brownfield expansion and greenfield projects is 75% for aspirational districts.
- Helping the country shoring up its much-needed healthcare infrastructure and creating more employment opportunities is one of the expected benefits. The main objective is to partially mitigate credit risk and facilitate bank credit at lower interest rates. **Hence, statement 1 is correct.**



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**4. (b)****Explanation:**

- A keystone species, ecosystem or process has a disproportionate influence on an ecosystem or landscape such as the role beavers play in altering the hydrological characteristics of streams and wetlands.
 - Keystone species have effects on biological communities that are disproportionate to their abundance and biomass.
 - The loss of keystone species results in broader community or ecosystem-level effects.
 - **A keystone species interacts with other species through predation, symbiotic dependencies such as plant-pollinator relationships, or ecosystem modification** (e.g., cavity nesters, beaver impoundments). **Hence, statement 1 is NOT correct.**
 - Wild salmon species are often considered an important keystone species as they add marine nitrogen to freshwater and terrestrial ecosystems and are an important food source for many animals, including bears that drag the carcasses of salmon into the forest, adding beneficial nitrogen into forest soils where it is limited.
 - Another example is sea otters that feed on sea urchins, which in turn feed on kelp. By limiting the number of sea urchins, the sea otter promotes the development of kelp forests which in turn provide habitat for fish and invertebrate species.
- A keystone ecosystem is particularly important because it provides habitat for a large portion or critical elements of an area's biodiversity.
 - Riparian ecosystems near streams, lakes and wetlands are considered keystone since they cover a relatively small area yet support a disproportionately large number of species.
 - Estuaries are also a keystone ecosystem because of their disproportionately large influence relative to their size and abundance. **Hence, statement 2 is correct.**

5. (a)**Explanation:****Greenwashing:**

- It is the practice of making an unsubstantiated or misleading claim about the environmental benefits of a product, service, technology or company practice. **Hence, option (a) is correct.**

- Greenwashing can make a company appear to be more environment friendly than it really is.

6. (d)**Explanation:**

- The **India-led United Nations Security Council (UNSC)** recently adopted Resolution 2593 on the Taliban. Sponsored by France, the UK and the US, it was adopted by 13 members, including India, voting in favour, none against it. While two permanent and veto-wielding members Russia and China abstained. The adoption of the resolution is a strong signal from the Security Council and the international community on its expectations in respect of Afghanistan. **Hence, statement 3 is correct.**
- **Reason of Abstention:**
 - **Russia:** The resolution wasn't specific enough about terror threats, did not speak of the "brain drain" effect of evacuating Afghans, and did not address the economic and humanitarian consequences of the US freezing the Afghan government's US accounts following the Taliban takeover.
 - **China:** Sharing the same concern as Russia, believes that the current chaos results from Western countries' "disorderly withdrawal". **Hence, statement 2 is correct.**
- **Resolution 2593 reiterates the importance of combating terrorism in Afghanistan**, including those individuals and entities designated according to resolution 1267 (1999).
- It called for the Taliban to facilitate safe passage for people wanting to leave Afghanistan, allow humanitarians to access the country, uphold human rights, including women and children, and be inclusive and negotiate a political settlement. **Hence, statement 1 is correct.**

7. (b)**Explanation:**

- **Convention on Migratory Species (Bonn Convention):** It is an environmental treaty of the United Nations which provides a global platform for the conservation and sustainable use of migratory animals and their habitats.
- **Ramsar Convention** is not affiliated with the United Nations systems of Multilateral Environmental Agreements (MEAs), **but it works very closely** with the other MEAs. **Hence, option (b) is correct.**



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- **Vienna Convention for protection of ozone layer:** The Convention was adopted by the Conference on the Protection of the Ozone Layer of the UN.
- **Rotterdam Convention:** It is an international treaty designed to facilitate informed decision making by countries with regard to trade in hazardous chemicals.

8. (c)

Explanation:

- **International Labour Organization (ILO)** is a specialised agency of the United Nations. It is the only tripartite UN agency. It brings together governments, employers, and workers of 187 member States to set labour standards, develop policies and devise programs promoting decent work for all women and men.
- Its published report titled **‘World Social Protection Report 2020–22’** has revealed that globally 4.1 billion people are living without any social safety net of any kind. **Hence, statement 1 is correct.**
- Social Security is a comprehensive approach designed to prevent deprivation, assure the individual of a basic minimum income for himself and his dependents, and protect the individual from any uncertainties. It includes access to health care and income security measures related to old age, unemployment, sickness, disability, work injury, maternity or the loss of the primary breadwinner in a family, and extra support for families with children.
- Global Population with Social Protection, Limited Unemployment Protection, Challenges Put Forward by Covid-19 Pandemic, Persisting Inequalities, Barriers to Accessing Healthcare are some of the highlights presented in the report. **Hence, statement 2 is correct.**

Steps taken:

- There is a need to recognise that adequate and comprehensive social protection is essential for social justice and decent work and creating a sustainable and resilient future.
- Establishing universal social protection and realising the human right to social security for all is the cornerstone of a human-centred approach to obtaining social justice.
- Collective financing, broad risk-pooling, and rights-based entitlements are necessary conditions for supporting adequate access to healthcare for all in a shock-responsive manner.
- Stronger linkages and better coordination between mechanisms for accessing medical care and income security are needed to address critical health determinants effectively.

9. (c)

Explanation:

- An alien plant also referred to as exotic, introduced, foreign, non-indigenous or non-native, is one that has been introduced by humans intentionally or otherwise through human agency or accidentally from one region to another.
- An alien plant that has escaped from its original ecosystem and is reproducing on its own in the **regional flora is considered a naturalized species.**
- Those naturalized aliens that become so successful as to **spread in the flora and displace native biota or threaten valued environmental**, agricultural or personal resources by the damage it causes are considered invasive. **Hence, statement 2 is correct.**
- According to the Convention for Biological Diversity, invasive alien species are the second largest cause of biodiversity loss in the world and impose high costs to agriculture, forestry, and aquatic ecosystems.
- In fact, introduced species are a greater threat to native biodiversity than pollution, harvest, and disease combined.
- Invasive species possess characteristic features like “pioneer species” in varied landscapes, tolerant of a wide range of soil and weather conditions, generalist in distribution, produces copious amounts of seed that disperse easily, grows aggressive root systems, short generation time, high dispersal rates, long flowering and fruiting periods, broad native range, abundant in native range.
- Invasive **species are likely to have relatively small amounts of DNA in their cell nuclei.**
- Apparently, the cells in these plants are able to divide and multiply more quickly and consequently the entire plant can grow more rapidly than species with higher cellular DNA content. This gives them a leg up in disturbed sites. **Hence, statement 1 is correct.**

10. (c)

Explanation:

- The Chief Minister of Manipur virtually launched the e-ILP platform to regulate the **Inner Line Permit (ILP) system** effectively.
- The ILP system came into effect in Manipur on 1st January 2020.
- **Four types of permits are issued - temporary, regular, special, and labour tickets.**



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- ILP is a document that Indian citizens from other states must possess to enter states like Arunachal Pradesh, Mizoram, Nagaland, and Manipur.
- It is issued by the concerned State Government and can solely be issued for travel purposes. **Hence statement 1 is correct.**
- Such states are exempted from the Citizenship Amendment Act (CAA) provisions.

Rules for Foreigners:

- Foreigners need a Protected Area Permit (PAP) to visit places different from ILPs required by domestic tourists. **Hence, statement 2 is correct.**
- Under the Foreigners (Protected Areas) Order 1958, all areas falling between the 'Inner Line', as defined in the said order, and the State's International Border are declared as a Protected Area.
- A foreign national usually is not allowed to visit a Protected/Restricted Area unless it is established to the Government's satisfaction that there are great reasons to justify such a visit.

11. (a)

Explanation:

Biomining:

- Biomining is the process of using microorganisms (microbes) to extract metals of economic interest from rock ores or mine waste. **Hence, option (a) is correct.**
- Biomining techniques may also be used to clean up sites that have been polluted with metals.
- It is usually used for old dumped waste that remains in a partly or fully decomposed state with no segregation in existence between wet and dry waste.
- In the cost-effective method of biomining, treatment is done by dividing the garbage heap at the site into suitable blocks to let the air percolate in the heap.
- As a result, the leachate which is the water in the heap with suspended solid particles is drained off and microbes are sprayed in the heap to initiate biological decompositions.
- The waste is turned over several times in order to devoid the waste to leachate as much as possible.
- This biological decomposition of the waste decreases the volume of the waste by 40%.

12. (d)

Explanation:

- **World Food Day** is celebrated to commemorate the establishment of the United Nations **Food and Agriculture Organisation (FAO)** on **16th October 1945**. Hence, **statement 1 is correct.**
- The Prime Minister of India released a **commemorative coin of Rs. 75 denomination** to mark the **75th Anniversary of the FAO**.
- FAO is a **specialised agency of the United Nations** that leads international efforts to defeat hunger.
- **About World Food Day:**
 - It is observed **annually on 16th October** to address the problem of global hunger.
 - It emphasises on the **Sustainable Development Goal 2 (SDG 2)** i.e. **Zero Hunger**.
 - **The theme for 2020:** "Grow, Nourish, Sustain. Together."
 - ◆ There has been a renewed focus on food, **nutrition**, health, immunity and sustainability due to the unprecedented challenges posed by the **Covid-19** pandemic.

75th Anniversary of FAO:

- Over the years, FAO has **helped to increase agricultural production and eradicate hunger all over the world**, including India and has played an important role in increasing nutrition.
- **2020's Nobel Peace Prize** to the **World Food Programme** is also a major achievement for the FAO as the programme was started by FAO.

The Desert Locust Programme 1952:

- The Desert Locust Programme was formally launched, following a request from an international conference convened by India. **Hence, statement 2 is correct.**
- This programme was greatly strengthened a decade later by cooperation with UNDP.
- The past seventy years have seen fundamental changes in FAO's strategy of locust control.
- Measures started out as defensive and were aimed at minimizing the damage in the aftermath of locust swarms.

UN Decade of Family Farming:

- The UN Decade of Family Farming 2019-2028 aims to shed new light on what it means to be a family farmer in a rapidly changing world and highlights more than ever before the important role they play in eradicating hunger and shaping our future of food.



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- The UN Decade of Family Farming is led by the Food and Agriculture Organization (FAO) and the International Fund for Agricultural Development (IFAD) in collaboration with governments and civil society organizations. **Hence, statement 3 is correct.**

13. (a)

Explanation:

- As an environmental treaty of the United Nations, Convention on the Conservation of Migratory Species of Wild Animals (CMS) provides a global platform for the conservation and sustainable use of migratory animals and their habitats. **Hence, statement 1 is correct.**
- CMS brings together the States through which migratory animals pass (the Range States) and lays the legal foundation for internationally coordinated conservation measures throughout a migratory range.
- As the only global convention specializing in the conservation of migratory species, their habitats and migration routes, CMS complements and co-operates with a number of other international organizations, NGOs and partners in the media as well as in the corporate sector.
- Migratory species threatened with extinction are listed on Appendix I of the Convention. **Hence, statement 2 is correct.**
- Migratory species that need or would significantly benefit from international co-operation are listed in Appendix II of the Convention.
- The agreements may range from legally binding treaties (called Agreements) to less formal instruments, such as Memoranda of Understanding, and can be adapted to the requirements of particular regions. **Hence, statement 3 is NOT correct.**
- As of 1 November 2019 the Convention on Migratory Species has 130 Parties. Two countries signed the original Convention but have yet to ratify it so are not a Party (Jamaica and the Central African Republic).

14. (c)

Explanation:

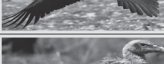
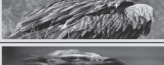
- World Meteorological Organization is a specialised agency of the United Nations. It released a report that stated that weather disasters killed 2 million in the last 50 years. **Hence, statement 1 is correct.**

- The report also warned that the failure to reduce disaster losses as set out in the 2015 Sendai Framework puts at risk the ability to develop countries to eradicate poverty and achieve other important Sustainable Development Goals (SDGs).
- **Sendai Framework 2015** was adopted at the Third United Nations World Conference on Disaster Risk Reduction in 2015 in Sendai, Miyagi, Japan. **Hence, statement 2 is correct.**
- The present Framework applies to the risk of small-scale and large-scale, frequent and infrequent, sudden and slow-onset disasters caused by natural or artificial hazards and related environmental, technological and biological hazards and risks.

15. (d)

Explanation:

- The following image describes the IUCN status of vulture species:

Sr. No.	Name of the Vulture Species	IUCN status	Pictorial Representation
1.	Oriental White-backed Vulture (Gyps Bengalensis)	Critically Endangered	
2.	Slender-billed Vulture (Gyps Tenuirostris)	Critically Endangered	
3.	Long-billed Vulture (Gyps Indicus)	Critically Endangered	
4.	Egyptian Vulture (Neophron Percnopterus)	Endangered	
5.	Red-Headed Vulture (Sarcogyps Calvus)	Critically Endangered	
6.	Indian Griffon Vulture (Gyps Fulvus)	Least Concerned	
7.	Himalayan Griffon (Gyps Himalayensis)	Near Threatened	
8.	Cinereous Vulture (Aegypius Monachus)	Near Threatened	
9.	Bearded Vulture or Lammergeier (Gypaetus Barbatus)	Near Threatened	

Hence, option (d) is correct.



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**16. (d)****Explanation:**

- **Botanic Gardens Conservation International (BGCI)** launched the State of the **World's Trees Report**.
- The report warns that **almost a third of the world's tree species are at risk of extinction**, while hundreds are on the brink of being wiped out.
- BGCI is a membership organisation representing botanic gardens in more than 100 countries worldwide. It is an **independent UK charity** established in **1987** to link the world's botanic gardens in a global network for plant conservation.
- The top three threats facing tree species are crop production, timber logging, and livestock farming, while climate change and extreme weather are emerging threats.
- At least 180 tree species are directly threatened by rising seas and severe weather, especially islands like magnolias in the Caribbean. **Hence, option (d) is correct.**
- Islands including St Helena (69% of trees threatened), Madagascar (59%) and Mauritius (57%) have highest proportion of threatened trees.
- The report offers five recommendations to bring species back from the brink and provides hope for future if conservation efforts continue.

17. (c)**Explanation:**

- The theory of island biogeography illustrates a basic principle that large areas usually contain more species than smaller areas with similar habitat because they can support larger and more viable populations. **Hence, statement 1 is correct.**
- The number of species on an island is determined by two factors: the distance from the mainland and island size.
- These would affect the rate of extinction on the islands and the level of immigration. Other factors being similar (including distance to the mainland), on smaller islands the chance of extinction is greater on larger ones.
- This is one reason why larger islands can hold more species than smaller ones.
- Energy from the sun, captured by plant photosynthesis, flows from trophic level to trophic level via the food chain.
- A trophic level is composed of organisms that make a living in the same way, that is they are all primary producers (plants), primary consumers (herbivores) or secondary consumers (carnivores). **Hence, statement 2 is correct.**

18. (c)**Explanation:****International Fund for Agricultural Development (IFAD):**

- IFAD is an international financial institution and specialized United Nations agency working in poverty eradication in the rural areas of developing countries, providing grants and loans with low interest for allied projects. **Hence, statement 1 is correct.**
- Established in 1977, it is the outcome of the World Food Conference of 1974. **Hence, statement 3 is NOT correct.**
- It works with marginalised and vulnerable groups such as farmers with small holdings, foresters, Pastoralists, Fishermen, and small-scale entrepreneurs by giving them disaster preparedness, access to weather information, technology transfer, and social learning.
- Its headquarters is in Rome, Italy. It has 177 member countries, of which India is also a member country. **Hence, statement 4 is NOT correct.**
- Recently, IFAD has committed to focusing 30% of its climate finance on supporting nature-based solutions in rural small-scale agriculture by 2030. It brings out the Rural Development Report every year. **Hence, statement 2 is correct.**

19. (c)**Explanation:****Petroleum Coke or Pet Coke:**

- Pet Coke is a dark solid carbon material and is an oil refinery by-product.
- It emits nearly 11 per cent **more greenhouse gases than coal**. **Hence, statement 1 is correct.**
- Pet coke is used as a **fuel because of its higher energy content than coal**, but it releases larger amounts of carbon dioxide and sulfur dioxide, which can cause lung disease and acid rain. **Hence, statement 2 is correct.**
- Recently India banned the import of pet coke for use as fuel due to rising pollution levels in major cities.
- **Import of pet coke is allowed for only cement, lime kiln, calcium carbide and gasification industries, when used as the feedstock or in the manufacturing process on actual user condition.**
- More than half of India's petcoke demand of 27 million tonnes is imported, mostly from the United States.

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**20. (c)****Explanation:**

- **Recreational forestry:** Raising of trees with the major objective of recreation alone. It is the practice of forestry with the object of raising flowering trees and shrubs mainly to serve as recreation forests for the urban and rural population.
- **Community forestry:** It is the raising of trees on community land and not on private land as in farm forestry. Community forestry was initially defined as, “any situation which intimately involves local people in a forestry activity.
- Planting of trees on the sides of roads, canals and railways, along with planting on wastelands is known as ‘**extension forestry**, increasing the boundaries of forests. Under this project there has been creation of woodlots in the village common lands, government wastelands and panchayat lands. **Hence, option (c) is correct.**

21. (d)**Explanation:**

- Recently, eight significant banks have joined the Account Aggregator (AA) network to enable customers to access and share their financial data easily.
- The RBI in 2016 approved AA as a new class of NBFC (Non-Banking Financial Companies), whose primary responsibility is to facilitate the transfer of users’ financial data with their explicit consent.
- An AA is a framework that facilitates sharing of financial information in a real-time and data-blind manner between regulated entities. **Hence, statement 1 is correct.**
- AAs enable the data flow between Financial Information Providers (FIPs) and Financial Information Users (FIUs). It permits users to control who gets access to their data, track and log its movement and reduce the potential risk of leakage in transit. **Hence, statement 2 is correct.**
- The architecture of AA is based on the Data Empowerment and Protection Architecture (DEPA) framework. It reduces the fraud associated with biological data by introducing secure digital signatures and end-to-end encryption for data sharing. It can also be extended to handle data from other domains. **Hence, statement 3 is correct.**

22. (d)**Explanation:**

- The Common Services Centres (CSC) has recently received approvals to manage and operate Passport Seva Kendra kiosks in rural areas. The Ministry of Electronics & IT (MeitY) initiative, the CSC, is part of the National e-Governance Plan (NeGP), approved by the Government in May 2006, as part of its commitment in the National Common Minimum Programme to introduce e-governance on a massive scale. **Hence, statement 2 is correct.**

The **objective of CSCs** is to provide-

- high quality and cost-effective videos.
- voice and data content and services in e-governance,
- education, health, telemedicine, and entertainment, including other private services.

The PPP (Public-Private Partnership) model of the CSC scheme envisages a **3-tier structure** consisting-

- CSC operator (called Village Level Entrepreneur or VLE).
- Service Centre Agency (SCA) that will be responsible for a division of 500-1000 CSCs.
- State Designated Agency (SDA) identified by the State Government responsible for managing the implementation in the entire State. **Hence, statement 1 is correct.**

CSC 2.0:

- It was launched in 2015, expanding the programme’s outreach to all gram panchayats. At least one CSC is envisaged in each 2.5 lakh Gram Panchayats.
- CSC 2.0 is a service delivery oriented entrepreneurship model with a large bouquet of services made available for the citizens through optimum utilisation of infrastructure already created in the form of State Wide Area Network (SWAN), State Service Delivery Gateway (SSDG), e-District, State Data Centre (SDC), and (National Optical Fiber Network (NOFN)/BharatNet. **Hence, statement 3 is correct.**

23. (d)**Explanation:****Western Ghats Ecology Expert Panel (WGEEP):**

- The **Western Ghats Ecology Expert Panel (WGEEP)**, also known as **Gadgil Committee**, and the **Kasturirangan Committee**, a **High-Level Working Group**, were constituted to **conserve and protect the biodiversity of Western Ghats** while allowing for sustainable and inclusive development of the region.

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- They recommended that **identified geographical areas falling in the six States of Kerala, Karnataka, Goa, Maharashtra, Gujarat and Tamil Nadu** should be declared as Ecologically Sensitive Area (ESA).
- A **draft notification related to the same was issued in 2018** mentioning the areas to be notified in the ESA.
- **Gadgil Committee:**
 - It recommended that **all of the Western Ghats should be declared as the ESA** with only limited development allowed in graded zones.
 - It classified the Western Ghats into **ESA 1, 2 and 3** of which **ESA-1 is a high priority zone** where almost all of the **developmental activities** (mining, thermal power plants, etc.) should be restricted.
 - It also recommended the **constitution of Western Ghats Ecology Authority (WGEA)**, as a statutory authority under the Ministry of Environment, Forests and Climate Change (MoEFCC) with the powers under **Section 3 of the Environment (Protection) Act, 1986**.
 - It was **criticised for being more environment-friendly and not in tune with the ground realities**.
- **Kasturirangan Committee:**
 - It sought to **balance the development and environment protection** in contrast to the system proposed by the Gadgil report.
 - **The committee's major recommendations were:**
 - ◆ Instead of the total area of Western Ghats, **only 37% of the total area to be brought under ESA**. Hence, **statement 1 is NOT correct**.
 - ◆ A complete **ban on mining, quarrying and sand mining** in ESA.
 - ◆ **No thermal power projects** to be allowed and **hydropower projects to be allowed only after detailed study**.
 - ◆ **Red industries** (highly polluting industries) to be strictly **banned**.
 - ◆ Exclusion of inhabited regions and plantations from the purview of ESAs making it a **pro-farmer approach**.

Ecologically Sensitive Areas:

- **Eco-Sensitive Zones or Ecologically Fragile Areas** are located **within 10 km of Protected Areas, National Parks and Wildlife Sanctuaries**. Hence, **statement 2 is NOT correct**.

- ESAs are **notified by the MoEFCC under Environment (Protection) Act 1986**.
- **Aim:** To regulate certain activities around National Parks and Wildlife Sanctuaries so as to minimise the negative impacts on the fragile ecosystem encompassing the protected areas.

24. (d)

Explanation:

- Recently, the Union Cabinet has approved the Production Linked Incentive (PLI) scheme for the textile sector.
- To boost domestic manufacturing and cut down on import bills, the union government in March 2020 introduced a PLI scheme that aims to give companies incentives on incremental sales from products manufactured in domestic units.
- The Scheme has also been approved for automobiles, pharmaceuticals, IT hardware including laptops, mobile phones & telecom equipment, white goods, chemical cells, food processing, etc. PLI for Textiles, RoSCTL, RoDTEP and other measures of Government in the sector, e.g. providing raw material at competitive prices, skill development, etc., will herald a new age in textiles manufacturing. It aims to promote the production of high-value Man-Made Fibre (MMF) fabrics, garments and technical materials. Hence, **statements 1 and 2 are correct**.
- **Expected Benefits:**
 - **Increase in Investment and Employment:**
 - ◆ It will lead to a new investment of more than Rs.19,000 crore, a cumulative turnover of over Rs. 3 lakh crore and additional employment opportunities of more than 7.5 lakh jobs in this sector and several lakhs more for supporting activities. Hence, **statement 3 is correct**.
 - ◆ The textiles industry predominantly employs women. Therefore, the scheme will empower women and increase their participation in the formal economy.
 - **Priority to Backward Areas:**
 - ◆ In addition, priority will be given to investment in Aspirational Districts, Tier 3, Tier 4 towns, and rural areas. Prioritising this will incentivise the Industry to move to backward regions.
 - ◆ This scheme will positively impact states like Gujarat, UP, Maharashtra, Tamil Nadu, Punjab, AP, Telangana, Odisha etc. Hence, **statement 4 is correct**.



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**25. (b)****Explanation:**

- India released National Action Plan on Climate Change (NAPCC) to mitigate and adapt to climate change.
- The action plan outlines a number of steps to simultaneously advance India's development and climate change-related objectives.
- The National Action Plan on Climate Change (NAPCC) encompasses a range of measures. It focuses on **eight missions**, which are as follows:
 - National Solar Mission
 - **National Mission for Enhanced Energy Efficiency**
 - National Mission on Sustainable Habitat
 - **National Water Mission**
 - **National Mission for Sustaining the Himalayan Ecosystem**
 - **Green India Mission**
 - National Mission for Sustainable Agriculture
 - National Mission on Strategic Knowledge for Climate Change. **Hence, option (b) is correct.**

26. (a)**Explanation:**

- **World Dugong Day** is celebrated on **May 28th** every year.
- **Dugong** (Dugong dugon) also called 'Sea Cow' is one of the four surviving species in the Order Sirenia and it is the **only existing species of herbivorous mammal that lives exclusively in the sea** including in India. **Hence, statement 1 is correct.**
- They are listed as **Vulnerable** on the IUCN Red List and are protected in India under **Schedule I** of the **Wild (Life) Protection Act, 1972**.
- According to a 2013 survey report of the Zoological Survey of India (ZSI), there were only about 200 dugongs in the **Gulf of Mannar in Tamil Nadu**, the **Andaman and Nicobar Islands** and the **Gulf of Kutch in Gujarat**. **Hence, statement 2 is NOT correct.**
- Dugongs are an **important part of the marine ecosystem** and their depletion will have effects all the way up the food chain. **Proper conservation** is the only way to save dugongs from extinction.

Threats:

- Dugongs **graze on seagrass** and the **loss of seagrass beds** due to **ocean floor trawling** is one of the most important factors behind decreasing dugong populations in many parts of the world.
- ◆ **Trawling is a method of fishing** that involves **pulling a fishing net through the water** behind one or more boats. It is **harmful to the environment** because it damages the seafloor, coral reefs and other marine animals.
- **Human activities** such as the destruction and modification of habitat, pollution, rampant illegal fishing activities, vessel strikes, unsustainable hunting or poaching and unplanned tourism are the main threats to dugongs.
- ◆ Dugong meat is consumed under the wrong impression that it cools down human body temperature.

Steps Taken for Conservation:

- In February 2020, India hosted the **13th Conference of Parties (CoP)** of the **Convention on the Conservation of Migratory Species of Wild Animals (CMS)**, an environmental treaty under the aegis of the **United Nations Environment Programme (UNEP)**.
- ◆ The Government of India is a **signatory to the CMS since 1983**.
- ◆ India has signed **non-legally binding Memorandums of Understanding (MoU)** with CMS on the conservation and management of **Siberian Cranes** (1998), **Marine Turtles** (2007), **Dugongs** (2008) and **Raptors** (2016).
- The **Ministry of Environment, Forests and Climate Change** constituted a '**Task Force for Conservation of Dugongs**' to look into issues related to conservation of dugongs and implementation of the '**UNEP/CMS Dugong MoU**' in India.
- ◆ It also facilitates **India to act as the leading nation in the South Asia Sub-region** with respect to dugong conservation.

27. (a)**Explanation:**

- Atmanirbhar Bharat Corner will be an exclusive space to promote GI (Geographical Indication) tagged tribal art and craft besides natural and organic products.

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- **Tribal Cooperative Marketing Development Federation of India (TRIFED), in collaboration with the Ministry of External Affairs, will set up an Atmanirbhar Bharat corner in 100 Indian Missions/ Embassies worldwide.**
- TRIFED is a national-level apex organisation functioning under the administrative control of the Ministry of Tribal Affairs. It is involved in schemes such as Van Dhan Programme, MSP for MFP and TRIFOOD.
- The Geographical Indication got recognition by the World Trade Organization (WTO). It denotes the geographical territory of the product and conveys an assurance of qualities or attributes that are unique to that specific geographic region.
- In India, the Geographical Indications (Registration and Protection Act), 1999, came into effect from September 2003. **Hence, option (a) is correct.**

28. (b)

Explanation:

According to the Central Water Commission (CWC) report on heavy metals contaminating Indian rivers:

- **Iron** emerged as the most common contaminant. **Hence, option (b) is correct.**
- **Arsenic** and **zinc** concentrations were found to be within the limits.
- Other major contaminants were **lead, nickel, chromium, cadmium and copper.**
- Metals found in the Non-Monsoon Period: Lead, cadmium, nickel, chromium and copper.
- The main sources of pollution are mining, milling, plating and surface finishing industries that discharge a variety of toxic metals into the environment.
- Long-term exposure to the heavy metals may result in physical, muscular, and neurological degenerative processes that mimic Alzheimer's disease, Parkinson's disease, muscular dystrophy and multiple sclerosis.

29. (c)

Explanation:

- The Brown to Green Report is the world's most comprehensive annual review of G20 climate action: it provides concise and comparable information on mitigation, finance and vulnerability. The report is published annually

by 'Climate Transparency', a global partnership of 14 climate research organisations and NGOs from the majority of G20 countries, many from emerging economies. Germanwatch is one of the main authors. **Hence, option (c) is correct.**

Findings:

- The G20 countries are responsible for around 80 % of global greenhouse gas emissions, and 85 % of global GDP. In the G20 countries, around 70 % of climate impacts could be avoided by limiting global warming to 1.5°C instead of 3°C. The G20 have a political responsibility as well as economic interest and capability to move the world towards a 1.5°C compatible pathway.
- Carbon emissions from the world's 20 biggest economies are rising. None of the G20 countries have plans that will put them on track to limit global warming to 1.5°C, despite the fact that most are technically capable and have economic incentives. To keep the Paris Agreement's 1.5°C goal within reach, G20 countries will have to increase their 2030 emission targets by 2020 and significantly scale up mitigation, adaptation and finance over the next decade.

30. (c)

Explanation:

- The Ministry of Commerce and Industry has revised the Transport and Marketing Assistance (TMA) scheme for specified agriculture products.
- It will be in effect from 1st April 2021 to 31st March 2022. It was launched in 2019 for **providing financial assistance for the transport and marketing of agricultural products to boost exports of such commodities to certain countries in Europe and North America.**
- The revised scheme has included dairy products, among other agricultural products in its purview and increased assistance rates. Rates of assistance have been improved, by 50% for exports by sea and 100% through the air.
- Other Objectives include assisting with the international component of freight and marketing of agricultural produce, mitigating the disadvantage of a higher cost of transportation of export of specified agriculture products due to trans-shipment and promoting brand recognition for Indian agricultural products in the selected overseas markets. **Hence, statement 1 is correct.**



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Agricultural and Processed Food Products Export Development Authority:

- It is a non-trading, statutory body established by the Government of India under the Agricultural and Processed Food Products Export Development Authority Act passed by the Parliament in December 1985.
- Functions under the Ministry of Commerce and Industry. The Authority has its headquarters in New Delhi.
- **It is responsible for export promotion and development of the scheduled products** viz. fruits, vegetables, meat products, dairy products, alcoholic and non-alcoholic beverages, etc. Another responsibility is to monitor the import of sugar. **Hence, statement 2 is correct.**

31. (d)

Explanation:

To curb the growing menace of air pollution, the National Clean Air Programme (NCAP) was launched by Ministry of Environment, Forest and Climate Change.

Goal:

- Goal of NCAP is to meet the prescribed annual average ambient air quality standards at all locations in the country in a stipulated timeframe. The tentative national level target of **20%–30% reduction of PM_{2.5} and PM₁₀ concentration by 2024** is proposed under the NCAP taking **2017** as the base year for the comparison of concentration. **Hence, statement 1 is NOT correct.**

Objectives:

- To augment and evolve effective and proficient ambient air quality monitoring network across the country for ensuring comprehensive and reliable database
- To have efficient data dissemination and public outreach mechanism for timely measures for prevention and mitigation of air pollution and for inclusive public participation in both planning and implementation of the programmes and policies of government on air pollution
- To have feasible management plan for prevention, control and abatement of air pollution.

Approach:

- Collaborative, Multi-scale and Cross-Sectoral Coordination between relevant Central Ministries, State Government and local bodies.
- Focus on no Regret Measures, Participatory and Disciplined approach.

- The NCAP will be a mid-term, five-year action plan with 2019 as the first year.
- **The plan aims to focus on 102 non-attainment cities identified for mitigation actions. Hence, statement 2 is NOT correct.**

32. (a)

Explanation:

- Rainforests are Earth's oldest living ecosystems, with some surviving in their present form for at least 70 million years. **Hence, statement 2 is NOT correct.**
- They are incredibly diverse and complex, home to more than half of the world's plant and animal species—even though they cover just 6% of Earth's surface.
- **Rainforests thrive on every continent except Antarctica. Hence, statement 1 is correct.**
- The largest rainforests on Earth surround the Amazon River in South America and the Congo River in Africa.
- The tropical islands of Southeast Asia and parts of Australia support dense rainforest habitats. Even the cool evergreen forests of North America's Pacific Northwest and Northern Europe are a type of rainforest.
- Rainforests' rich biodiversity is incredibly important to our well-being and the well-being of our planet. Rainforests help regulate our climate and provide us with everyday products.

33. (c)

Explanation:

- The revamped US-India Strategic Clean Energy Partnership SCEP was recently launched during the ministerial meeting of the Petroleum and Natural Gas Ministry with the US Ministry of Energy.
- Another pillar, Emerging Fuels, was commenced in the present four pillars of the SCEP. With this, the SCEP inter-governmental engagement now spreads across five pillars of cooperation - Power and Energy Efficiency, Responsible Oil and Gas, Renewable Energy, Sustainable Growth and Emerging Fuels. **Hence, statement 1 is correct.**
- Support India in achieving its goal of installing 450GW of renewable energy capacity by 2030. A new India-U.S. Task Force on Biofuels was also announced.
- The SCEP was launched under the US - India Climate and Clean Energy Agenda 2030 Partnership announced by both countries at the Leaders' Summit on Climate held earlier this year (2021). **Hence statement 2 is correct.**



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**34. (b)****Explanation:**

- The first National Wildlife Action Plan (NWAP) was adopted in 1983. It outlined the strategies and action points for Wildlife conservation. It was revised by the Wildlife Action Plan (2002-2016). **Hence, statement 1 is NOT correct.**
- The National Afforestation Programme was launched in 2002, involving plantation in degraded forests of the country. **Hence, statement 2 is correct.**

35. (d)**Explanation:**

- The **Great Indian Bustard, Asian Elephant and Bengal Florican** have been included in Appendix I of UN Convention on Migratory Species at the 13th Conference of the Parties (COP) to the Convention on Migratory Species (CMS), Gandhinagar (Gujarat). **Hence, option (d) is correct.**
- India's proposal to include all the three species in the Appendix I was unanimously accepted by the 13th COP to the CMS.
- A migratory species may be listed in Appendix I provided that the best scientific evidence available indicates that the species is endangered.

Asian Elephant:

- India is the natural home of the largest population of Asian elephants. It is also found in Nepal, Bangladesh, Bhutan and Myanmar.
- It usually resides in shrublands, artificial/terrestrial forests and grasslands.
- It is listed as '**Endangered**' on the **IUCN Red List** of threatened species. It is also listed in **Appendix I** of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) and **Schedule I** of the Wildlife (Protection) Act, 1972.
- The challenges confronting Asian elephant conservation in most elephant Range States are habitat loss and fragmentation, human-elephant conflict, and poaching and illegal trade of elephants.

Great Indian Bustard:

- The Great Indian Bustard is one of the **heaviest flying birds** in the world.

- It usually resides in dry grasslands and scrublands on the Indian subcontinent; its largest populations are found in the Indian state of Rajasthan.
 - The Great Indian Bustard is the state bird of Rajasthan.
- It is listed as '**Critically Endangered**' on the **IUCN Red List**. It is also listed in **Appendix I** of CITES and **Schedule I** of the Indian Wildlife (Protection) Act, 1972.
- 90% of its population has been reduced within 50 years (six generations) majorly due to poaching.

Bengal Florican:

- The species has two disjunct populations, one in the Indian Subcontinent, the other in South-East Asia. The former occurs in Indian Subcontinent mainly in India (Uttar Pradesh, Assam and Arunachal Pradesh.) and terai region of Nepal.
- It inhabits lowland dry, or seasonally inundated, natural and semi-natural grasslands, often interspersed with scattered scrub or patchy open forest.
- It has been listed as '**Critically Endangered**' on the **IUCN Red List**. The bird is listed under **Schedule I** of the Wildlife Protection Act of India, 1972 and **Appendix I** of CITES.
- It has a very small, rapidly declining population largely as a result of widespread loss of its grassland habitat.

36. (d)**Explanation:**

- **Swachh Survekshan Grameen 2021:**
 - Recently, the Jal Shakti Ministry launched Swachh Survekshan Grameen 2021 or rural cleanliness survey 2021 under Swachh Bharat Mission (Grameen) Phase-II.
 - It is carried out by an expert agency that assesses the state of hygiene, cleanliness and sanitation in rural India as a part of the Centre's initiative to award Open Defecation Free (ODF) Plus status to villages. **Hence, statements 1 and 2 are correct.**
- **Swachh Bharat Mission (Grameen) Phase-II:** emphasises the sustainability of achievements under phase I and provides adequate facilities for Solid/Liquid & plastic Waste Management (SLWM) in rural India. It will be implemented from 2020-21 to 2024-25 in a mission mode with a total outlay of Rs. 1,40,881 crores. **Hence, statement 3 is correct.**

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- **Swachh Bharat Mission** was launched on 2nd October 2014 to accelerate the efforts to achieve universal sanitation coverage and is being implemented by the Ministry of Housing and Urban Affairs (M/o HUA) and the Ministry of Jal Shakti for urban and rural areas, respectively.
 - Under the mission, all villages, Gram Panchayats, Districts, States, and Union Territories in India declared themselves “Open-Defecation Free” (ODF) by 2nd October 2019, the 150th birth anniversary of Mahatma Gandhi, by constructing over 100 million toilets in rural India. It further led to Jal Jeevan Mission (JJM) addressing water security.

37. (c)

Explanation:

- The **Convention on Biological Diversity (CBD)** is a legally binding multilateral environmental agreement that has 194 contracting Parties (Countries) as its members with three objectives – conservation of biological diversity, sustainable use of the diversity and ensuring fair and equitable sharing of benefits of such use. **Hence, statement 1 is correct.**
- It entered into force on 29th December 1993.
- The CBD provides sovereign rights over their biological resources, and requests countries would facilitate access to the genetic resources by other parties subject to national legislation and on mutually agreed terms. **Hence, statement 2 is correct.**
- The CBD also provides for equitable sharing of benefits arising from the utilization of traditional knowledge, and practices, with holders of such knowledge.
- **India's biodiversity in comparison with the World:**
 - India is one of the 17-mega biodiversity countries of the world. With only 2.4% of the land area, India already accounts for 7-8% of the recorded species of the world.
 - Over 46,000 species of plants and 81,000 species of animals have been recorded in the country so far by the Botanical Survey of India and the Zoological Survey of India, respectively. India is an acknowledged centre of crop diversity, and harbors many wild relatives and breeds of domesticated animals and fish besides millions of microbial diversity, insects and other species.

● **Biodiversity Act - 2002:**

- The Biodiversity Act - 2002 primarily addresses issues of conservation, sustainable use of biological resources in the country, issues related to access to genetic resources and associated knowledge and fair and equitable sharing of benefits arising from utilization of biological resources to the country and its people.
- The Act and the Rules are implemented in India through a decentralized system.
- A three tiered structure has been established under the Act at the national, state and local levels.
 - ◆ At the local level, the Biodiversity Management Committees (BMCs) are to be established by institutions of local self-government for implementation of specific provisions of the Act and Rules.
 - ◆ At the state level, the State Biodiversity Boards (SBBs) are established to deal with all matters relating to implementation of the Act and the Rules.
 - ◆ At the national level, the National Biodiversity Authority (NBA) is established to deal with all matters relating to implementation of the Act and the Rules.
 - ◆ Each of these structures are required to be connected for decision making processes on various issues, including on issues of access and benefit sharing (ABS).

38. (c)

Explanation:

● **Colour- Coded Weather Warning:**

- It is issued by the Indian Meteorological Department (IMD) whose objective is to alert people ahead of severe or hazardous weather which has the potential to cause damage, widespread disruption or danger to life.
- Warnings are updated daily.
- The IMD uses 4 colour codes are:
 - ◆ **Green (All is well):** No advisory is issued.
 - ◆ **Yellow (Be Aware):** Yellow indicates **severely bad weather spanning across several days**. It also suggests that the weather could change for the worse, causing disruption in day-to-day activities. **Hence, statement 1 is correct.**



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- ◆ **Orange/Amber (Be prepared):** The orange alert is issued as a warning of extremely bad weather with the potential of disruption in commute with road and rail closures, and interruption of power supply. Hence, statement 2 is correct.
- ◆ **Red (Take Action):** When the extremely bad weather conditions are certainly going to disrupt travel and power and have significant risk to life, the red alert is issued.
- These alerts are universal in nature and are also issued during floods, depending on the amount of water rising above land/in a river as a result of torrential rainfall.
 - For instance, when the water in a river is 'above normal' level, or between the 'warning' and 'danger' levels, a yellow alert is issued.

39. (d)

Explanation:

- The **Insurance Regulatory and Development Authority of India (IRDAI)** has issued revised guidelines for trade credit insurance.
- **Trade Credit Insurance:**
 - It protects businesses against the risk of non-payment for goods and services. Hence, statement 4 is correct.
 - It usually covers a portfolio of buyers and indemnifies an agreed percentage of an invoice or invoices that remain unpaid due to protracted default or insolvency.
 - It contributes to economic growth by facilitating trade and helps improve financial stability by addressing trade losses because of payment risks. Hence, statement 1 is correct.
 - It can be issued to sellers or suppliers of goods or services, factoring companies defined in the Factoring Regulation Act, 2011 and banks and financial institutions.
 - Banks, financial institutions, and factoring companies cover the loss on account of non-receipt of payment from a buyer, due to commercial or political risks, against the bills and invoices purchased or discounted.
- ◆ **Commercial risks** include insolvency or extended default of the buyer, rejection by the buyer after delivery subject to contract conditions, and rejection before shipment and non-receipt of payment on account of the collecting bank's failure.

- ◆ **Political risk** cover is available only in the case of buyers outside India. It will include the war between the buyer's country and India and war, hostilities, civil war, rebellion, revolution, insurrection, or other disturbances in the buyer's country.
- These guidelines will apply to all insurers transacting general insurance business registered under the Insurance Act, 1938. However, ECGC Ltd (formerly Export Credit Guarantee Corporation of India Ltd) is exempted from applying these guidelines. Hence, statement 2 is correct.
- It will facilitate general insurance companies to help businesses manage country risk, open up access to new markets and manage non-payment risk associated with trade financing portfolios.
- It will also enable general insurance companies to offer trade credit insurance with customised covers to improve businesses for the Micro, Small, and Medium Enterprises (MSMEs), considering these enterprises' evolving insurance risk needs. Hence, statement 3 is correct.

40. (b)

Explanation:

- **Satkosia Tiger Reserve, Bhubaneswar** comprises two adjoining sanctuaries of central Odisha named as **Satkosia Gorge Sanctuary and Baisipalli Sanctuary**.
- Satkosia spreads along the gorge over the mighty river Mahanadi in Odisha. Hence, statement 1 is NOT correct.
 - These two sanctuaries together covering an area of 963.87 sq km were notified as **Tiger Reserve in December 2007**.
- Lying in a **transitional zone** extending between the **Chhota Nagpur Plateau and the Deccan Plateau**, the tiger reserve exhibits endemic life forms of both biotic provinces. Hence, statement 2 is correct.
- **Flora and Fauna:**
 - The area supports **moist deciduous forest, dry deciduous forest and moist peninsular Sal forest**.
 - This area is the home for **Tiger, Leopard, Elephant, Gaur, Chousingha, Sloth bear, Wild dog**, varieties of resident and migratory birds, reptilian species, etc.



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● **Satkosia Gorge Sanctuary:**

- It owes its name to the **narrow stretch of River Mahanadi** i.e. “Sat-Kosh” or seven miles long near **Tikarpada**, 60 km south of Angul.
- The area was **made a sanctuary in 1976** and spreads out in four districts of Odisha namely Angul, Budh, Cuttack and Nayagarh.
- The Satkosia Gorge is a **unique feature in geomorphology of India** because here **Mahanadi cuts right across the Eastern Ghats** and has formed a magnificent gorge.
- **Fauna:** It is known for Gharials, Mugger crocodile and rare freshwater turtles like Chitra indica and Indian softshell turtle.

● **Baisipalli Sanctuary:**

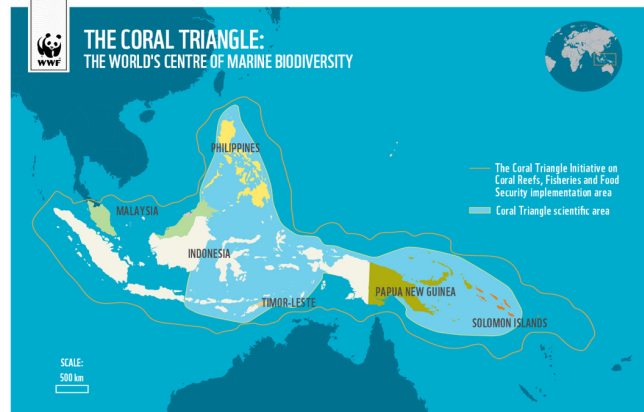
- It gets its name from the **22 settlements existing within its span**.
- It was given the status of **sanctuary in May 1981**.
- It is located where the **Mahanadi River passes through a gorge in the Eastern Ghats** mountains in **Nayagarh District**.
- The whole area is a part of the **Deccan Peninsula Biogeographic Zone, Eastern Plateau province and Eastern Ghat sub-division**.
- **Flora and Fauna:** It is a **sal dominated forest** with a significant number of tigers, leopards, elephants, herbivores like Chousingha and water birds, reptilians,

41. (b)

Explanation:

Coral Triangle:

- The Coral Triangle is a marine area located in the western Pacific Ocean. It includes the waters of Indonesia, Malaysia, the Philippines, Papua New Guinea, Timor Leste and Solomon Islands. **Hence, statement 1 is NOT correct.**
- The Coral Triangle also supports large populations of commercially important tuna, fueling a multi-billion dollar global tuna industry. Over 120 million people live in the Coral Triangle and rely on its coral reefs for food, income and protection from storms.



- Corals bleach, or lose their bright colors, in water that is too warm. They die in water that isn't salty enough. Without the reef structure, the ecosystem collapses. **Hence, statement 2 is correct.**
- Most coral reef ecosystems will bounce back from collapse. As ocean temperature cools and retains more salt, the brightly colored corals return. Slowly, they build reefs. Algae, plants, and animals also return.

42. (d)

Explanation:

- The updated Groundswell report released by the World Bank indicated that climate change could force 216 million people across six world regions to move within their countries by 2050. **Hence, statement 1 is correct.**
- It has two reports whose combined findings provide, for the first time, a global picture of the potential scale of internal climate migration across the six World Bank regions.
- **First Groundswell Report:** It was published in 2018 and used a robust and novel modelling approach to understand the scale, trajectory, and spatial patterns of future climate migration within countries, focusing on three regions: Sub-Saharan Africa, South Asia, and Latin America.
- **Second Groundswell Report:** It builds on the first report, applying the same approach to three new regions: the Middle East and North Africa, East Asia and the Pacific, and Eastern Europe and Central Asia. **Hence, statement 2 is correct.**
- **Policy Recommendations:**
 - **Reduce Emission:**
 - ◆ Five years after the Paris Agreement, the world is still headed for at least three °C of warming by 2100.



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- ◆ Ambitious action to curb global emissions is critical to reducing the burden of climate change impacts essential resources, livelihood systems, and urban centres that may drive people to migrate in distress.
- **Inclusive and Resilient Development Pathways:** Integrating internal climate migration in development planning is critical to address the poverty factors that make people particularly vulnerable to climate change impacts, such as a lack of viable livelihood options and lower-quality assets.
- **Plan for Each Migration Phase:** Planning for internal climate migration means accounting for all migration phases - before, during, and after moving. Before migration, adapt-in-place solutions can help communities stay in a place where local adaptation options are viable and sensible.
 - ◆ During migration, policies and investments can enable mobility for people who need to move away from unavoidable climate risks.
 - ◆ After migration, planning can ensure that both sending and receiving areas are well equipped to meet the needs and aspirations of their populations.
- **Investments:** More investments are needed in research at scale, including new, more granular data sources and differentiated climate change impacts, to better contextualise and understand internal climate migration at the regional and country level. **Hence, statement 3 is correct.**
- **The United Nations Framework Convention on Climate Change (UNFCCC) formally incorporates mobility in the context of climate change in the 2010 Cancun Adaptation Framework.**
- It's "measures to enhance understanding, coordination and cooperation concerning climate-induced displacement, migration, and planned relocation" while "taking into account their common but differentiated responsibilities". **Hence, statement 4 is correct.**
- There are 161 UNESCO Global geoparks spread across 44 countries. At present India does not have any Global Geopark. **Hence, statement 2 is NOT correct.**
- **The four features that are fundamental to a UNESCO Global Geopark are:**
 - **Geological heritage of international value:**
 - ◆ In order to become a UNESCO Global Geopark, the area must have a geological heritage of international value.
 - **Management:**
 - ◆ UNESCO Global Geoparks are managed by a body having legal existence recognized under national legislation.
 - **Visibility:**
 - ◆ UNESCO Global Geoparks promote sustainable local economic development mainly through geotourism.
 - ◆ In order to stimulate the geotourism in the area, it is crucial that a UNESCO Global Geopark has visibility.
 - **Networking:**
 - ◆ A UNESCO Global Geopark is not only about cooperation with the local people living in the UNESCO Global Geopark area, but also about cooperating with other UNESCO Global Geoparks through the Global Geoparks Network (GGN), and regional networks for UNESCO Global Geoparks, in order to learn from each other and, as a network, improve the quality of the label UNESCO Global Geopark.
 - ◆ Working together with international partners is the main reason for UNESCO Global Geoparks to be a member of an international network such as the GGN.

43. (a)

Explanation:

- **UNESCO Global Geoparks** are single, unified geographical areas where sites and landscapes of international geological significance are managed with a holistic concept of protection, education and sustainable development. **Hence, statement 1 is correct.**

44. (c)

Explanation:**Components of Water Analysis:**

- **Biochemical Oxygen Demand-**
 - BOD is the amount of dissolved oxygen used by microorganisms in the biological process of metabolizing organic matter in water. **Hence, statement 2 is correct.**

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- The more organic matter there is (e.g., in sewage and polluted bodies of water), the greater the BOD; and the greater the BOD, the lower the amount of dissolved oxygen available for higher animals such as fishes.
- The BOD is, therefore, a reliable gauge of the organic pollution of a body of water.
- One of the main reasons for treating wastewater prior to its discharge into a water resource is to lower its BOD—i.e., reduce its need for oxygen and thereby lessen its demand from the streams, lakes, rivers, or estuaries into which it is released.
- **Chemical Oxygen Demand-**
 - COD is a method of estimating how much oxygen would be depleted from a body of receiving water as a result of bacterial action.
- **Coliforms-**
 - Water pollution caused by fecal contamination is a serious problem due to the potential for contracting diseases from pathogens (disease-causing organisms).
 - The presence of pathogens in the water is determined with indirect evidence by testing for an “indicator” organism such as coliform bacteria. **Hence, statement 1 is correct.**
 - Coliforms are bacteria that are always present in the digestive tracts of animals, including humans, and are found in their wastes. They are also found in plant and soil material.
- **Presence of any heavy metals like Arsenic, Phosphorus, etc.**
 - Heavy metals are known to induce cardiovascular diseases, developmental abnormalities, neurologic and neurobehavioral disorders, diabetes, hearing loss, hematologic and immunologic disorders.
 - Studies have shown that toxic heavy metals can interfere with the absorption and use of nutritionally essential metals such as iron, calcium, copper, and zinc.

45. (a)**Explanation:**

- In India, yellow rust is a major disease in the Northern Hill Zone and the North-Western Plain Zone and spreads easily during the onset of cool weather and when wind conditions are favourable.
 - Rain, dew and fog favour the disease’s development.
- Hence, option (a) is correct.**

- Yellow Rust disease appears as yellow stripes of powder or dust on leaves and leaf sheaths of the wheat crop.
 - This yellow powder comes out on clothing or fingers when touched.
 - The disease can spread rapidly under congenial conditions and affects crop development, and eventually the yield.
- Yield due to the disease can be affected by between 5 and 30 per cent.
 - This occurs when the rust colonies in the leaves drain the carbohydrates from the plant and reduce the green leaf area.
- In 2019, a new variety of wheat called HD-3226 or Pusa Yashasvi was released by the Indian Agricultural Research Institute, which had higher levels of resistance against major rust fungi such as the yellow/stripe, brown/leaf and black/stem.

46. (c)**Explanation:**

- **Deepfakes** are modified images, text, audio, and video or synthetic media created with the help of Artificial Intelligence. **Hence, statement 1 is correct.**
- It generates a fake version from original or authentic audio-visual content by superimposing new audio or image on an existing media file. Media is manipulated with such sharpness that it becomes almost impossible to identify the difference between fake and tangible media. It can only be determined by using AI tools.

Threats of Deepfakes:

- It can be used to disrupt democratic processes like elections in any country.
- They are used to stain the reputations of individuals and spread propaganda against them.
- According to a **deepfake tracking research organisation**, over 100,000 fake nude images of women were circulating online.
- Terrorist organisations and insurgents can use these technologies to further their agenda of destabilising state governments. They can spread false information about institutions, public policy, and politicians for this purpose.

Regulations of Deepfakes:

- **In the US:** As per US law, Social Media Companies cannot be held responsible for the posts on their platforms.

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- **In India:** Some provisions in the Indian Penal Code (IPC) and Information Technology Act, 2000 criminalise a few related cybercrimes. But there is no specific law to deal with deep fakes. The existing legal framework of many countries, including India, does not criminalise deep fakes. **Hence, statement 2 is correct.**

47. (d)

Explanation:

Tundra:

- Northern and Southernmost region of the world adjoining the ice bound poles.
- **Devoid of trees except stunted shrubs** in the southern part of tundra biome, **ground flora includes lichen, mosses and sedges.** The typical animals are reindeer, arctic fox, polar bear, snowy owl, lemming, arctic hare, and ptarmigan. Reptiles and amphibians are almost absent.

Taiga:

- The dominating vegetation is coniferous evergreen mostly spruce, with some pine and firs. **Hence, statement 1 is NOT correct.**

Grasslands:

- Grassland biomes consist of large open areas of grass. Trees can be present, but they are infrequent. Low rainfall, wildland fires, and grazing by animals are three factors that maintain grasslands. **Hence, statement 2 is NOT correct.**
- In grassland regions, the climate is ideal for the growth of grasses only. The low precipitation rates are enough to nourish grasses but not enough for a forest of trees. Frequent fires also play a role in maintaining grassland ecosystems.
- Grasses are well adapted to grow back after a fire. Grassland animals are also prepared for fires, fleeing or burrowing underground to wait out the flames.
- Large animals, such as African elephants, can also trample the ground and discourage the growth of trees.

48. (d)

Explanation:

- The Union Cabinet, in September 2021, has approved the Prime Minister POSHAN scheme or PM-POSHAN for providing hot cooked meal in Government and Government-aided schools. The scheme will cover 11.8 crore students enrolled in classes 1 to 8 in 11.2 lakh schools.

Primary (1-5) and upper primary (6-8) schoolchildren are currently entitled to 100 grams and 150 grams of food grains per working day each to ensure a minimum of 700 calories.

- The scheme will replace the existing national program for mid-day meals in schools or the **National Programme Mid-Day Meal Scheme.** It has been launched for an initial period of five years (2021-22 to 2025-26). **Hence, statement 1 is correct.**
- The new scheme provides supplementary nutrition for children in the **aspirational districts** and those prone to **anaemia.** **Hence, statement 2 is correct.**
- The scheme will be extended to students studying in **pre-primary or Balvatikas** running in government and government-aided primary schools (Balvatika is the pre-school started in government schools last year to include children younger than six years in the formal education system). **Hence, statement 3 is correct.**
- Involvement of the **Farmers Producer Organizations (FPO) and Women Self Help Groups** in implementing the scheme will be encouraged. Use of locally grown traditional food items for a fillip to local economic growth, i.e. **Vocal for Local**, will be enabled. **Hence, statement 4 is correct.**

49. (b)

Explanation:

- The government has launched a **Global Environment Facility (GEF)** assisted project namely, "Green – Ag: Transforming Indian Agriculture for global environment benefits and the conservation of critical biodiversity and forest landscapes" in collaboration with the **Food and Agriculture Organisation (FAO).** **Hence, option (b) is correct.**
- It is being implemented in high-conservation-value landscapes of five States namely (i) Madhya Pradesh : Chambal Landscape, (ii) Mizoram: Dampa Landscape, (iii) Odisha: Similipal Landscape, (iv) Rajasthan: Desert National Park Landscape and v) Uttarakhand: Corbett-Rajaji Landscape.
- The project seeks to mainstream biodiversity, climate change and sustainable land management objectives and practices into Indian agriculture.
- The overall objective of the project is to catalyze transformative change of India's agricultural sector to support achievement of national and global environmental benefits and conservation of critical biodiversity and forest landscapes.



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- The project will support harmonization between India's agricultural and environmental sector priorities and investments so that the achievement of national and global environmental benefits can be fully realized without compromising India's ability to strengthen rural livelihoods and meet its food and nutrition security.
- **Global Environment Facility (GEF):**
 - The Global Environment Facility (GEF) was established on the eve of the 1992 Rio Earth Summit to help tackle our planet's most pressing environmental problems.
 - In combination with its traditional investments under the Conventions, the GEF is:
 - ◆ Strategically focusing its investments to catalyze transformational change in key systems that are driving major environmental loss, in particular energy, cities and food.
 - ◆ Prioritizing integrated projects and programs that address more than one global environmental problem at a time, building on the GEF's unique position and mandate to act on a wide range of global environmental issues.
 - ◆ Implementing new strategies and policies to enhance results, including stronger engagement with the private sector, indigenous peoples, and civil society, and an increased focus on gender equality.

50. (b)**Explanation:**

- The Union Cabinet has approved capital infusion in the Export Credit Guarantee Corporation (ECGC) and its listing through an initial public offering.
- The government will inject **Rs 4,400 crore** in the ECGC for five years, beginning 2021-22. It will enable it to expand its coverage to export-oriented industries, particularly labour-intensive sectors. Listing ECGC on the stock market is also being initiated to raise more funds. **Hence, statement 1 is NOT correct.**
- The Cabinet also approved continuing the National Export Insurance Account (NEIA) scheme and infusion of Rs 1,650 crore Grant-in-Aid over five years. **Hence, statement 2 is correct.**
- National Export Insurance Account (NEIA) Scheme:
 - NEIA Trust was **established in 2006** to promote project exports from India that are of strategic and national importance.

- It promotes Medium and Long Term (MLT)/project exports by extending (partial/full) support to covers issued by ECGC to MLT/project export.
- Exim Bank, in April 2011, in conjunction with ECGC Ltd., introduced a new initiative, viz. Buyer's Credit under the NEIA scheme, under which the Bank finances and facilitates project exports from India.

51. (b)**Explanation:**

- Climate variability and climate change are being felt all over the Himalayan region, which is one of the most sensitive climate zones globally.
 - Receding apple lines, changing cropping patterns, more disasters like landslides and floods, drying springs in hills, spread of vector-borne diseases are the impacts related to climate variability and climate change, felt all over the Himalayan region.
- Indian scientists have developed a common framework for assessment of climate change vulnerability in all the states in the Himalayan region - **Assam, Manipur, Meghalaya, Mizoram, Nagaland, Tripura, Arunachal Pradesh, Sikkim, Himachal Pradesh, Uttarakhand, Jammu and Kashmir and hilly districts of West Bengal.**
 - An index based on socio-economic factors, demographic and health status, sensitivity of agricultural production, forest-dependent livelihoods and access to information, services and infrastructure has been developed.
- Latest assessment shows that the vulnerability index is the **highest for Assam (0.72)** and Mizoram (0.71), followed by Jammu & Kashmir (0.62), Manipur (0.59), Meghalaya and West Bengal (both 0.58), Nagaland (0.57), Himachal Pradesh and Tripura (0.51 both), Arunachal Pradesh (0.47) and Uttarakhand (0.45). **Hence, statement 1 is NOT correct.**
- Sikkim is the least vulnerable state with the index being 0.42.
- Assam is highly vulnerable to climate change because of factors like **low per capita income**, deforestation, **large number of marginal farmers**, least area under irrigation, lack of alternative sources of income and high rates of poverty.
 - All of these make people vulnerable to climate change as they have less capacity to adapt. **Hence, statement 2 is correct.**

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- The vulnerability assessments are developed so that it provides useful information to the officials, decision makers, funding agencies and experts to have a common understanding on vulnerability and enable them to plan for adaptation.
- The assessment has been done jointly by experts from Indian Institutes of Technology at Guwahati and Mandi, in collaboration with Indian Institute of Science, Bangalore with support from the Department of Science and Technology (DST) and the Swiss Development Corporation (SDC) which is implementing the Indian Himalayas Climate Adaptation Program (IHCAP).

52. (c)

Explanation:

- INSPIRESat-1 CubeSat, has been developed under the **International Satellite Program in Research and Education (INSPIRE)**.
- It will be placed in a low earth orbit, equipped with a **Compact Ionosphere Probe (CIP)** for **studying the earth's ionosphere**.
 - The ionosphere is the part of the atmosphere ionised by solar radiation.
 - CIP is an all-in-one plasma sensor that uses a single instrument to perform multiple sensor functions in a time-sharing mechanism.
- It will also provide a greater understanding of why the **Sun's corona** is hotter than the photosphere, changes during different solar events, and how these events affect the earth's ionosphere.
- It weighs less than 10kg and will be **launched aboard an upcoming Polar Satellite Launch Vehicle (PSLV) of the Indian Space Research Organisation (ISRO)**. **Hence, statement 1 is correct.**
 - PSLV is the third generation launch vehicle of India. It is the first Indian launch vehicle to be equipped with liquid stages.
 - It is a four-stage launch vehicle with the first and third stages using solid rocket motors and the second and fourth stages using liquid rocket engines.
- The INSPIRE is a consortium of global space universities formed to advance science and engineering. The INSPIRE program aims to provide a constellation of Earth and space-weather observing satellites. **Hence, statement 2 is correct.**

- The following organisations developed it:
 - Laboratory for Atmospheric and Space Physics (LASP) at the University of Colorado Boulder in the U.S.
 - National Central University, Taiwan
 - Nanyang Technological University in Singapore
 - Indian Institute of Space Science and Technology (IIST)

53. (c)

Explanation:

The Future of Earth, 2020 Report:

- The South Asia Future Earth Regional Office, the Divecha Centre for Climate Change and the Indian Institute of Science have released **"The Future of Earth, 2020" report. Hence, statement 1 is correct.**
- The report has been prepared with the **aim** of reducing carbon footprint and halting global warming below 2 degree Celsius by 2050. **Hence, statement 2 is correct.**

Key Findings:

- **Five Global Risks**
 - Climate change
 - Extreme weather
 - Biodiversity loss
 - Food crisis
 - Water crisis
- **Example of Interrelation between different Global Risks–**
 - Extreme heat waves can accelerate global warming by releasing large amounts of stored carbon from affected ecosystems, and at the same time intensify water crises and/or food scarcity.
 - The loss of biodiversity weakens the capacity of natural and agricultural systems to cope with climate extremes, thus increasing vulnerability to food crises.
 - Strains on food production are expected to increase, as a result of various forces including climate change, biodiversity loss, and a global population on the rise.
- **Greenhouse Gas Emissions–**
 - Despite declarations of a climate crisis or climate emergency by the leaders of more than 700 cities, states and governments, the concentration of carbon dioxide in the atmosphere during 2019 reached more than 415 parts per million (ppm).
 - The five years from 2014 to 2018 were the warmest recorded over land and ocean since 1880.



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● Decline in Biodiversity–

- Humans have significantly altered 75% of the planet's land area. About a quarter of species in assessed plant and animal groups are threatened.
- In 2018, the world's last male northern white rhino died in his Kenyan enclosure, while the Brazilian blue parrot, Spix's Macaw, was declared extinct in the wild.

54. (c)

Explanation:

- Features of the Drone Rules, 2021 (Notified on 25 August 2021) are:
 1. Based on a premise of trust, self-certification, and non-intrusive monitoring.
 2. Several permissions and approvals were abolished. The number of forms was reduced from 25 to 5. Types of fees reduced from 72 to 4.
 3. Digital sky platform is being developed as a user-friendly online single-window system.
 4. Interactive drone airspace map with red and yellow zones.
 5. No permission is required for operating drones in green zones.
 6. The yellow zone, where ATC permission is required, has been reduced from 45 km to 12 km from the airport perimeter. **Hence, statement 3 is correct.**
 7. No remote pilot licenses required for micro drones (for non-commercial use).
 8. No security clearances required before the issuance of any registration license.
 9. Under Drone Rules, 2021 coverage of drones increased from 300 kg to 500 kg. This will cover drone taxis also. **Hence, statement 1 is NOT correct.**
 10. No restriction on foreign ownership in Indian drone companies.
 11. No requirement for import clearance from DGCA.
 12. Remote pilot license to be issued by DGCA within 15 days of the pilot receiving the remote pilot certificate from an authorised drone school.
 13. The maximum penalty for violations is reduced to INR 1 lakh. It was several lakhs earlier. **Hence, statement 2 is correct.**

14. Drone corridors will be developed for cargo deliveries.
15. Drone promotion council to be set up by the government with participation from academia, startups, and other stakeholders.

55. (c)

Explanation:

- The smallest member of the pig family, the pygmy hog, is a **critically endangered species**.
- Once found along a narrow strip of tall and wet grassland plains on the Himalayan foothills – from Uttar Pradesh to Assam, through Nepal's terai areas and Bengal's duars – it was thought to have become extinct in the 1960s. But **in 1971 it was "re-discovered"** with a small population in the **Bornadi Wildlife Sanctuary**.
- The only viable population of the pygmy hog in the wild is in the Manas Tiger Reserve in Assam. **Hence, statement 1 is correct.**
- The pygmy hog is one of the very few mammals that build its own home, or nest, complete with a 'roof'.
- It is an indicator species, its presence reflects the health of its primary habitat, the tall, wet grasslands of the region. One of the main reasons for its plummeting numbers was grassland degradation.

Recovery Programme for Critically Endangered Species

- The programme is one of the three components of the centrally funded scheme, **Integrated Development of Wildlife Habitats (IDWH)**.
- Started in 2008-09, IDWH is meant for providing support to protected areas (national parks, wildlife sanctuaries, conservation reserves and community reserves except tiger reserves), protection of wildlife outside protected areas and recovery programmes for saving critically endangered species and habitats.
- So far, **22 species have been identified under the recovery programme**. These are the Snow Leopard, Bustard (including Floricans), Dolphin, Hangul, Nilgiri Tahr, Caracal, Marine Turtles, Dugongs, Edible Nest Swiftlet, Asian Wild Buffalo, Nicobar Megapode, Manipur Brow-antlered Deer, Vultures, Malabar Civet, Indian Rhinoceros, Asiatic Lion, Swamp Deer, Jerdon's Courser, the Northern River Terrapin, Clouded Leopard, Arabian Sea Humpback Whale and Red Panda. **Pygmy Hog is not a part of this programme. Hence, statement 2 is correct.**



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**56. (b)****Explanation:****National Disaster Management Authority (NDMA):**

- NDMA is the **apex statutory body** for disaster management in India. **Hence, statement 1 is NOT correct.**
 - The Prime Minister of India heads it. **Hence, statement 2 is correct.**
 - It was formally constituted on **27th September 2006 by the Disaster Management Act, 2005.**
 - Its primary purpose is to coordinate response to natural or artificial disasters and capacity-building in disaster resiliency and crisis response.
 - It envisages building a safer and disaster-resilient India by a holistic, proactive, technology-driven, and sustainable development strategy that involves all stakeholders and fosters a culture of prevention, preparedness, and mitigation.
- NDMA has been implementing a central sector scheme, namely **Aapda Mitra**, since May 2016, focusing on training 6000 community volunteers (200 volunteers per district) in disaster response in selected 30 most flood-prone districts of 25 States of India. **Hence, statement 3 is correct.**
 - The scheme aims to provide the community volunteers with the skills needed to respond to their community's immediate needs in the aftermath of a disaster, enabling them to undertake essential relief and rescue tasks during emergencies such as floods, flash floods, and urban flooding.
 - Based on positive feedback and strong recommendations of States/UTs for the training of more volunteers in the **hazard-prone districts of India**, NDMA is also planning to upscale the scheme in pan India to train 100,000 Volunteers in **350 selected highly vulnerable sections** prone to flood, cyclone, landslide, and earthquake across the country. This would be of great help to the community. This will also promote a culture of volunteerism in DRR in India.

57. (d)**Explanation:**

- The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 provides for the restitution of deprived forest rights across India, in-

cluding both individual rights to cultivated land in forest-land and community rights over common property resources.

- Nodal Agency for the implementation is Ministry of Tribal Affairs. **Hence, statement 1 is NOT correct.**
- The maximum limit of the recognizing rights on forest land is 4 hectares. **Hence, statement 2 is NOT correct.**
- National Parks and Sanctuaries have been included along with Reserve Forest, Protected Forests for the recognition of Rights. **Hence, statement 3 is correct.**

58. (c)**Explanation:**

- On September 27, 2021, India successfully test-fired an upgraded variant of its Akash surface-to-air missile (SAM).
- The upgraded missile, named Akash Prime, intercepted and destroyed an unmanned aerial target in a flight test.
- It was developed by **Defence Research & Development Laboratory (DRDL)**, Hyderabad, collaborating with **other DRDO laboratories under the Missiles and Strategic Systems (MSS)**. The Akash system entered Indian Army service in 2009.
- Compared to the existing Akash system, it is equipped with an active indigenous **RF (Radio Frequency)** seeker for improved accuracy, ensuring that the missile's target is hit.
- Other improvements were also incorporated in Akash Prime, like ensuring reliable performance under a low-temperature environment at higher altitudes. **Hence, statement 1 is correct.**

Akash Missile:

- Akash is India's **first indigenously produced medium-range SAM** to engage multiple targets from multiple directions. It can be launched from mobile platforms like battle tanks or wheeled trucks. It has a nearly **90% kill probability.**
- The DRDO started the development of the Akash SAM in the late 1980s as part of the **Integrated Guided Missile Development Programme**. **Hence, statement 2 is correct.**
- It is unique in that It can simultaneously engage multiple targets in the group or autonomous modes.

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- It has built-in **Electronic Counter-Counter Measures (ECCM) features**, which means that it has mechanisms on-board that can counter the electronic systems that deceive the detection systems.
- The missile is supported by the indigenously developed radar called '**Rajendra**'.
- It can engage targets at a speed of **2.5 times more than the speed of sound** and detect and destroy targets flying at low, medium, and high altitudes.
- The missile is reportedly cheaper and more accurate than US Patriot missiles due to its solid-fuel technology and high-tech radars.

59. (d)

Explanation:

Cartagena Protocol:

- The Cartagena Protocol on Biosafety is an additional agreement to the Convention on Biological Diversity (CBD). **Hence, statement 1 is correct.**
 - It establishes procedures for regulating the import and export of LMOs from one country to another. **Hence, statement 2 is correct**
 - It also requires Parties to ensure that Living Modified Organisms (LMOs) being shipped from one country to another are handled, packaged and transported in a safe manner.
 - The shipments must be accompanied by documentation that clearly identifies the LMOs, specifies any requirements for the safe handling, storage, transport and use and provides contact details for further information.
- The Cartagena Protocol has provisions for an **Advance Informed Agreement (AIA)** procedure:
 - Under the AIA procedure, a country intending to export an LMO for intentional release into the environment must notify in writing to the Party of import before the first proposed export takes place. **Hence, statement 3 is correct.**
- India is a party to the Cartagena Protocol (ratified in 2003). **Hence, statement 4 is correct.**
 - The nodal agency (Competent National Authority-CNA) in the country for the implementation of the Protocol is the Ministry of Environment, Forest and Climate Change (MOEF&CC), Government of India.

60. (a)

Explanation:

- **Judima, a homemade rice wine of Assam's Dimasa tribe**, has become the first traditional brew in the northeast to bag a Geographical Indication (GI) tag. This is the second product from the hill districts of **Karbi Anglong** and **Dima Hasao** to get the GI tag.
 - Judima is a rice wine made from sticky rice (glutinous rice called **Bora**), steamed and mixed with traditional herbs called **Thembra (Acacia pennata)**.
 - The wine is a speciality of the **Dimasa tribe** of the state (Assam) and has a distinct sweet taste. It takes around one week to prepare and can be stored for years.
 - There are around 14 recognised Plain Tribe communities, 15 Hills Tribe communities, and 16 recognised Scheduled Caste communities in the state.
 - The Bodos are the largest group, comprising almost half of the state's tribal population. Other major ST groups include Missing, Karbi, Rabha, Kachari, Lalung, and Dimasa.
 - The **pokkali** variety is known for its **saltwater resistance** and flourishes in the rice paddies of the **coastal Alappuzha, Ernakulam, and Thrissur districts of Kerala**. The single-season paddy is raised in saltwater fields between June and November, followed by a season of fish farming. The uniqueness of the rice has brought it the Geographical Indication (GI) tag and is the subject of continuing research.
 - On September 29, 2021, it was informed by the state government that Madhya Pradesh had been recognised in the claim of Madhya Pradesh and Maharashtra for GI tag to **Chinnor rice**. Chinnor rice is mainly produced in the **Balaghat region**, although it is also produced in Bhandara, Maharashtra, for which Maharashtra claimed the GI tag. The **chalky loamy soil** found in Balaghat creates favourable conditions for rice production there. That is why Balaghat is also called the **rice bowl of Madhya Pradesh**.
 - Other Recent GI Tagged Products from Assam:
 - Kaji Nemu (a type of lemon) (2020)
 - **Chokuwa Rice** of Assam (2019)
- Hence, option (a) is correct.**



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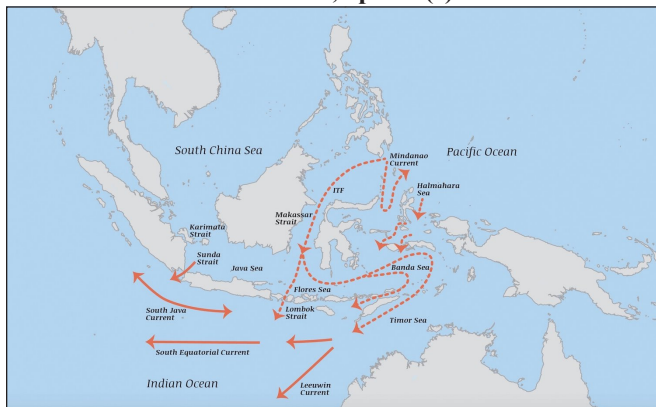
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61. (d)**Explanation:**

- An important characteristic of all communities is that their composition and structure constantly change in response to the changing environmental conditions.
- This change is orderly and sequential, parallel with the changes in the physical environment.
- These changes lead finally to a community that is in near equilibrium with the environment and that is called a climax community.
- The gradual and fairly predictable change in the species composition of a given area is called ecological succession.
- During succession some species colonise an area and their populations become more numerous, whereas populations of other species decline and even disappear.
- The entire sequence of communities that successively change in a given area are called sere(s). **Hence, option (d) is correct.**
- The individual transitional communities are termed seral stages or seral communities. In the successive seral stages there is a change in the diversity of species of organisms, increase in the number of species and organisms as well as an increase in the total biomass.

62. (c)**Explanation:**

- The third edition of the maritime exercise 'Samudra Shakti' between **India and Indonesia** was conducted from **20th-22nd September 2021** in the **Sunda Strait**. In pursuance of India's Act East Policy, Exercise 'Samudra Shakti' was conceived in 2018 as a **bilateral IN-INDN** exercise. **Hence, option (c) is correct.**



- Other Exercises with Indonesia:
 - Garuda Shakti (Military Exercise)
 - IND-INDO CORPAT,
 - IND-INDO BILAT (Maritime Exercise)

63. (a)**Explanation:**

- Keoladeo National Park, Rajasthan and Loktak Lake, Manipur have been included in Montreux Record in 1990 and in 1993 respectively.
- **Chilika Lake, Orissa** included in **Montreux Record** in 1993 was removed in November 2002. It was placed on the Montreux Record due to problems caused by siltation and sedimentation which was choking the mouth of the lake. **Hence, option (a) is correct.**
- Chilika Lake was removed from the Record in 2002 following rehabilitation efforts for which the Chilika Development Authority received the Ramsar Wetland Conservation Award for 2002.

64. (b)**Explanation:****3rd State Food Safety Index:**

- The index was developed by **FSSAI** (Food Safety and Standards Authority of India) to measure states' performance on five significant food safety parameters.
- The parameters include **Human Resources and Institutional Data, Compliance, Food Testing – Infrastructure and Surveillance, Training & Capacity Building, and Consumer Empowerment.**
- The Index is a dynamic quantitative and qualitative benchmarking model that provides an objective framework for evaluating food safety across all States/UTs.
- The first State Food Safety Index for 2018-19 was announced on the first-ever World Food Safety Day on 7th June 2019.
- Further, based on the fact that similar States should be compared, FSSAI for generation of SFSI Index has classified States/ UTs into **3 categories (Large States, Small States, and Union Territories)**. This will ensure comparability among similar entities.
- Gujarat was the top-ranking state among the larger states, followed by Kerala and Tamil Nadu.
- Among the smaller states, **Goa stood first**, followed by Meghalaya and Manipur. **Hence, option (b) is correct.**
- UTs, Jammu & Kashmir, Andaman & Nicobar Islands, and New Delhi were secured top ranks.

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65. (b)

Explanation:**Global Tiger Forum:**

- The **Global Tiger Forum (GTF)** is the **only inter-governmental international body** established with members from willing countries to embark on a global campaign to protect the Tiger. **Hence, statement 1 is correct.**
- Utilizing co-operative policies, common approaches, technical expertise, scientific modules and other appropriate programmes and controls the GTF is focused on saving the remaining 5 subspecies of Tigers distributed over 13 Tiger Range countries of the world.
- The GTF was formed in **1993** on recommendations from an international symposium on Tiger Conservation at New Delhi, India.
- The GTF has a General Assembly meeting every 3 years and Standing committee meetings at least once a year.
- A Chairperson, usually a Minister from one of the Tiger Range countries heads GTF for a fixed tenure of 3 Years. **Hence, statement 2 is NOT correct.**
- The Secretariat of GTF is headed by a Secretary General and is located in **New Delhi, India. Hence, statement 3 is correct.**

66. (b)

Explanation:**Global Innovation Index (GII):**

- The **GII** was launched by the **World Intellectual Property Organization (WIPO)**, a specialised agency of the United Nations. The **GII's** overall formula for measuring an **economy's innovative capacity** and output clarifies decision-makers in government, business, and elsewhere as they look forward to creating policies that enable their people to invent and develop more efficiently. **Hence, statement 1 is correct.**
- The 2021 edition of the Global Innovation Index (GII) presents the latest global innovation ranking of **132 economies**, relying on **81 different indicators**. While tracking the most recent global innovation trends in the new Global Innovation Tracker, this edition also focuses on the impact of the COVID-19 pandemic on innovation. **Hence, statement 2 is correct.**
 - The **Global Innovation Tracker** is a new feature of the Global Innovation Index 2021 that captures key innovation trends. It uses a set of indicators identified in the **three stages of the innovation journey**: Science and Innovation Investment, Technological Progress, and Socioeconomic Impacts.

- The world's most innovative economy in 2021 is **Switzerland**, followed by Sweden, the United States of America (U.S.), the United Kingdom (U.K.), and the Republic of Korea, according to the **GII 2021** rankings. **Hence, statement 3 is NOT correct.**

- **India has climbed two spots and has been ranked 46th** in the Global Innovation Index (GII) 2021 rankings.

- Indicators: The index ranks world economies according to their innovation capabilities and consists of 81 indicators grouped into innovation inputs and outputs.

- **Innovation inputs:** Institutions; Human capital and research; Infrastructure; Market sophistication; Business sophistication.

- **Innovation outputs:** Knowledge and technology outputs; Creative outputs

- **India's Performance:**

- India has been on an upward trajectory over the past few years in the **GII**. India has shot up from 81 in 2015 to 46 in 2021. **Hence, statement 4 is correct.**

- India will perform better in innovation outputs than innovation inputs in 2021.

- This year India ranks 57th in innovation inputs, the same as last year but higher than 2019.

- As for innovation outputs, India ranks 45th. This position is the same as last year but higher than in 2019.

- India ranks 2nd among the 34 lower-middle-income group economies.

- India ranks 1st among the 10 economies in Central and Southern Asia.

- The government attributed the country's improved performance to atomic energy, science, and technology, biotechnology, and space.

67. (b)

Explanation:

- Eucalyptus trees are prone to fire. The leaves of these contain a highly inflammatory oil that ignites easily. The trees catch fire as the ground beneath the trees is usually littered with leaves. **Hence, statement 1 is NOT correct.**

- Forest fires are caused by both anthropogenic as well as natural reasons. In India, however, over 95 % of fire incidents are of anthropogenic origin.

- **Forest fires are generally of three types:**

- **Surface fire:** Surface fire is a very fast moving fire, which consumes small vegetation and surface litter along with loose debris. This type of forest fires are mostly common in India. **Hence, statement 2 is correct.**

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- **Crown Fire:** Crown fire advances from top to top of trees or shrubs without any close link with surface fire. It is the fastest way of spreading fire and most destructive for trees as well as wildlife. It is rarely observed in India.
- **Ground fire:** A true ground fire is not easily predictable as it spreads within, rather than top of organic matter. It consumes organic matter like duff, musk or peat present beneath the surface litter of the forest floor. It has the unique characteristic of having a smouldering edge with no flame and little smoke. Ground fires are most hard to tackle. There should be proper policy and practices to control these types of Fires.

68. (c)

Explanation:

- The “Three Sisters” recognized by the World Trade Organization’s (WTO) Sanitary and Phytosanitary Measures (SPS) Agreement are **International Plant Protection Convention (IPPC) with the Codex Alimentarius Commission for food safety standards and the World Organization for Animal Health (OIE)** for animal health standards. **Hence, option (c) is correct.**
- Food and agricultural standards are essential to efficient global food trade. Standards about quality characteristics, safety, process and labeling are important for judging conformity and authenticity at each step along the chain from production, through processing to commercial exchanges.
- The rules of the World Trade Organization often are the foundation for these current standards for plants, animals and food and the push for harmonization.
- The Three Sisters are the Codex Alimentarius Commission, the World Organization for Animal Health and the international and regional organizations operating within the framework of the International Plant Protection Convention.

69. (c)

Explanation:

- The 2021 Global Agricultural Productivity Report (GAP Report) urges the acceleration of productivity growth at all production scales to meet consumers’ needs and address current and future threats to human and environmental well-being.

- **Total Factor Productivity (TFP)** is growing at an annual rate of **1.36% (2020-2019)**. This is below the Global Agricultural Productivity Index that has set a yearly target of 1.73% growth to sustainably meet the needs of consumers for food and bioenergy in 2050. **Hence, statement 1 is correct.**
- TFP growth is influenced by climate change, weather events, changes in fiscal policy, market conditions, investments in infrastructure, and agricultural research and development. Human-caused climate change has slowed global agricultural productivity growth by 21 per cent since 1961. In drier regions of Africa and Latin America, climate change has slowed productivity growth by 34 per cent. (Ortiz-Bobea et al., 2021)
- Middle-income countries, including India, China, Brazil, and the countries of the former USSR, continue to have the most robust TFP growth rates.
- Low-income countries, home to many small-scale farmers, have a negative TFP growth rate of -0.31 per cent annually. **Hence, statement 2 is correct.**
- Nearly all agricultural output growth in low-income countries comes from land-use change, the destruction of forests and grasslands for cultivation and grazing.
- **Way forward:**
 - **Maximising agriculture’s climate change mitigation potential** is essential for sustainability, yet for most of the world’s producers, adapting to climate change and protecting their livelihoods is the most immediate challenge.
 - Small and large farms can be equally efficient with access to various **productivity-enhancing inputs, agronomic knowledge, and markets, producers of all scales** can optimise their productive potential. (Fuglie et al., 2019)
 - Given its proven effectiveness in boosting productivity and economic growth, investments in public-sector agricultural research and extension need significant increases. (FAO, 2018)

70. (c)

Explanation:

Weathering:

- Weathering describes the **breaking down or dissolving of rocks** and minerals on the surface of the Earth. Water, ice, acids, salts, plants, animals, and changes in temperature are all agents of weathering.



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- Weathering and erosion constantly change the rocky landscape of Earth. Weathering wears away exposed surfaces over time.

Mechanical Weathering:

- Mechanical weathering, also called physical weathering and disaggregation, causes rocks to crumble.
- **Water:** Water, in either liquid or solid form, is often a key agent of mechanical weathering. For instance, liquid water can seep into cracks and crevices in rock. If temperatures drop low enough, the water will freeze. When water freezes, it expands. The ice then works as a wedge. It slowly widens the cracks and splits the rock.
- **Temperature:** Temperature changes can also contribute to mechanical weathering in a process called **thermal stress**. Changes in temperature cause rock to expand (with heat) and contract (with cold). As this happens over and over again, the structure of the rock weakens. Over time, it crumbles.
- **Pressure:** Changes in pressure can also contribute to exfoliation due to weathering. In a process called **unloading**, overlying materials are removed. The underlying rocks, released from overlying pressure, can then expand. **As the rock surface expands, it becomes vulnerable to fracturing in a process called sheeting.**
- **Haloclasty:** Salt also works to weather rock in a process called haloclasty. Saltwater sometimes gets into the cracks and pores of rock. If the saltwater evaporates, salt crystals are left behind. As the crystals grow, they put pressure on the rock, slowly breaking it apart. **Hence, statement 1 is correct.**

Chemical Weathering:

- The action of a set of chemical processes operating at the atomic and molecular levels to break down and reform rocks and minerals.
E.g. When exposed to water anhydrite transforms to gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) **Hence, statement 2 is correct.**

71. (b)

Explanation:

Aquatic organisms are classified on the basis of their zone of occurrence and their ability to cross these zones:

- **Neuston:**
 - These are unattached organisms which live at the air water interface such as floating plants, etc. **Hence, option (b) is correct.**

- Some organisms spend most of their lives on top of the air-water interface such as water striders, while others spend most of their time just beneath the air-water interface and obtain most of their food within the water. E.g., beetles and back-swimmers

- **Periphyton:** These are organisms which remain attached to stems and leaves of rooted plants or substances emerging above the bottom mud such as sessile algae and their associated group of animals.
- **Plankton:** This group includes both microscopic plants like algae (phytoplankton) and animals like crustaceans and protozoans (zooplankton) found in all aquatic ecosystems, except certain swift moving waters.
- **Nekton:** This group contains animals which are swimmers. The nektons are relatively large and powerful as they have to overcome the water currents.
- The animals range in size from the swimming insects (about 2 mm long) to the largest animals, the blue whale.
- **Benthos:** The benthic organisms are those found living in the bottom of the water mass.

72. (b)

Explanation:

Rail Kaushal Vikas Yojana:

- Rail Kaushal Vikas Yojna is one of the **skill enhancement training programs** under the **Ministry of Indian Railways and Skill Development and Entrepreneurship**.
- The Ministry of Railways launched Rail Kaushal Vikas Yojana (RKVY) under **Pradhan Mantri Kaushal Vikas Yojana (PMKVY)**. **Hence, statements 1 and 2 are correct.**
- The program is for the unemployed youths of India to **impart technical training** in various trades to enhance their employability and entrepreneurship. **Free-of-cost training** is being imparted in Railway Training Institutes for eligible youths across the country.
- Any Matric passed student between 18-35 years age limit can select and apply for training online and offline.
- One candidate will be allowed to avail of training in only one trade and only one time. No repetitive courses will be allowed to any candidate so that other candidates may also get a chance. Since it is a short duration course, 75% attendance is a must to continue the training and issue of a certificate.
- There will be a test after completion of training, and certificates will be issued only to successful candidates.



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- There will be **no reservation** given to the candidates under this Scheme. The training will be conducted irrespective of religion, caste, creed, or race. **Hence, statement 3 is correct.**
- Candidates who get training under this scheme will have **no claim to seek employment** in Railways based on training. **Hence, statement 4 is NOT correct.**

73. (b)

Explanation:

- Recently, the International Energy Agency (IEA) released the World Energy Outlook (WEO) Report 2021. **Hence, option (b) is correct.**
- Published every year, the WEO provides critical analysis and insights on energy demand and supply trends. The 2021 report signalled pressure on governments to push for greater climate action at the Conference of Parties (COP26) summit (in Glasgow, UK).

Highlights:

- In 2020, even while economies bent under the weight of Covid-19 lockdowns, renewable sources of energy such as wind and solar PV continued to multiply, and electric vehicles set new sales records.
- At the moment, however, every data point showing the speed of change in energy can be countered by another showing the stubbornness of the status quo.
- This special edition of the *World Energy Outlook* has been designed to assist decision-makers at the 26th Conference of the Parties (COP26) and beyond by describing the critical decision points that can move the energy sector onto safer ground. It provides a detailed stocktake of how far countries have come in their clean energy transitions, how far they still have to go to reach the **1.5 °C goal** and the actions that governments and others can take to seize opportunities and avoid pitfalls way.
- The two pledges are:
 - **Stated Policies Scenario (STEPS):**
 - ◆ This looks at the measures and policies governments have already put in place. Despite the efforts, annual worldwide emissions would still be the same as developing countries build up their infrastructure.

- ◆ Under this scenario, temperatures in 2100 would be 2.6 C higher than pre-industrial levels.

○ Pledge for Net Zero:

- ◆ This looks at governments' pledges to achieve net-zero emissions, potentially doubling clean energy investment over the next decade.
- ◆ If countries manage to implement these pledges in time, the global average temperature increase would be **around 2.1 C by 2100** — an improvement, but still well above the **1.5 Celsius agreed under the Paris accord.**

74. (d)

Explanation:

Skinks:

- It is the **largest family of lizards**, having long bodies, relatively small or no legs, no pronounced neck and glossy scales. **Hence, statement 3 is correct.**
- It is found around homes, garages, and open spaces such as parks and school playgrounds, and around lakes.
- Skinks are **non-venomous, highly alert, agile and fast-moving and actively forage for a variety of insects and small invertebrates.**
- They have a **prominent role in maintaining ecosystems**, however, **not much is known about their breeding habits, and ecology** because identification of the species can be confusing.
- In India, **skinks are found in all kinds of habitats**, from the Himalayas to the coasts and from dense forests to the deserts. **Hence, statement 1 is correct.**
- **The Study:**
 - Recently, the Zoological Survey of India (ZSI) published a study on skinks (a kind of lizard), titled Skinks of India.
 - It gives a **phylogenetic** (evolutionary development) and **bio-geographical analysis** of the distribution of these species in all the **11 bio-geographic zones of India.**
 - India is **home to less than 4% of the skinks** across the globe. **62 species of skinks** are found in India and about 57% of them (**33 species**) are **endemic. Hence, statement 2 is correct.**



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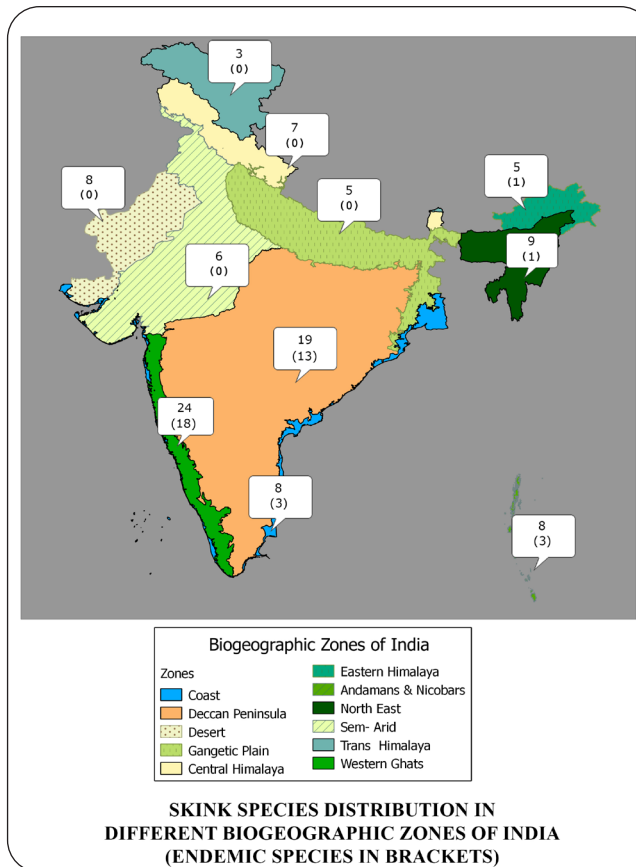
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75. (d)

Explanation:**Bannerghatta National Park:**

- The Bannerghatta National Park is located near **Bangalore in Karnataka**.
- Wildlife such as elephants, gaur, leopard, jackal, fox, wild boar, sloth, Sambar, Chital, spotted deer, barking deer, common langur, bonnet macaque, porcupine, and hares are found in abundance.
- The Bannerghatta Biological Park has been an integral part of Bannerghatta National Park.
- The **Eco-Sensitive Zone (ESZ) Expert Committee** of the Ministry of Environment and Forests (MOEF), has recommended the **reduction of nearly 100 sq. km** in the ESZ of Bannerghatta National Park.
- ESZ provides a regulated buffer zone around protected areas.

Pobitora Wildlife Sanctuary:

- It has the highest density of **one-horned rhinos** in the world and second highest concentration is in **Assam** after Kaziranga National Park.
- It is often called '**Mini Kaziranga**' due to similar landscape and vegetation.
- The wildlife sanctuary is home to endangered one-horned rhinoceros and the other mammals such as Leopard, Leopard cat, Fishing cat, Jungle cat, Feral Buffalo, Wild pigs, Chinese pangolins, etc.
- About 72% of Pobitora Sanctuary consists of wet savannah of *Arundo donax* and *Saccharum*. The remaining area is covered by water bodies.

Silent Valley National Park is in Kerala.**Mukurthi National Park:**

- It is a protected area located in the northwest corner of **Tamil Nadu** in the Western Ghats.
- The park was created to protect its keystone species, the **Nilgiri Tahr**.
- As a part of Western Ghats it is also a part of the **UNESCO World Heritage Site**.
- The park is characterised by **montane grasslands** and shrublands interspersed with **sholas** in a high altitude area of high rainfall, near-freezing temperatures and high winds.
- It is also home to an array of endangered wildlife, including royal Bengal tiger and Asian elephant, but its main mammal attraction is the Nilgiri tahr.
- It is part of **Nilgiri Biosphere Reserve** along with **Mudumalai Wildlife Sanctuary, Bandipur National Park, Nagarhole National Park, Wayanad Wildlife Sanctuary and Silent Valley**.
- Recently, Nilgiri tahr's population has increased from 568 in 2018 to 612 in 2019 in the Mukurthi National Park, Tamil Nadu.

Hence, option (d) is correct.

76. (b)

Explanation:

- NITI Aayog, has launched the Shoonya Campaign on September 15, 2021. **Hence, statement 2 is correct.**
- It is an initiative to promote **zero-pollution delivery vehicles** by working with consumers and the industry. The campaign aims to accelerate the **adoption of electric vehicles (EVs) in the urban deliveries segment** and create consumer awareness about the benefits of zero-pollution delivery. **Hence, statement 1 is NOT correct.**

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- **Shoonya Brand:** A corporate branding and certification program is being launched to recognise and promote the industry's efforts towards transitioning to EVs for final-mile deliveries. It will help e-commerce companies to distinguish their offerings from those of their competitors.
- **Online Tracking Platform:** An online tracking platform will share the campaign's impact through data such as vehicle kilometres electrified, carbon savings, criteria pollutant savings, and other benefits from clean delivery vehicles.

77. (d)

Explanation:

- Recently, scientists have unravelled the mystery behind Odisha's 'Black Tigers' in Similipal Tiger Reserve (STR).
- It is the only tiger habitat globally with melanistic tigers, which have broad black stripes running across their bodies and are thicker than those seen on normal tigers. **Hence, option (d) is correct.**
- Black Tigers are a rare colour variant of the tiger and are not a distinct species or geographic subspecies.
- The coat colouration and patterning that makes the wild cats appear dark boil down to a single mutation in the Transmembrane Aminopeptidase Q (Taqppe) gene.
- Due to Geographic Isolation, genetically related individuals have been mating for many generations in Similipal, leading to inbreeding.
- It should be noted that this has important implications for tiger conservation as such isolated and inbred populations are prone to extinction over even short periods.

Similipal Tiger Reserve:

- It was formally designated a tiger reserve in 1956 and brought under **Project Tiger in 1973**. It was declared a biosphere reserve by the Government of India in June 1994.
 - It has been part of the **UNESCO World Network of Biosphere Reserve since 2009**.
 - It is part of the Similipal-Kuldiha-Hadgarh Elephant Reserve, popularly known as Mayurbhanj Elephant Reserve, which includes three protected areas, i.e. Similipal Tiger Reserve, Hadagarh Wildlife sanctuary, and Kuldiha wildlife sanctuary.
- It is situated in the northern part of Odisha's **Mayurbhanj district**. Geographically, it lies in the eastern end of the east ghat.

- Similipal is home to a wide range of wild animals, including tigers and elephants, besides 304 birds, 20 species of amphibians, and 62 species of reptiles.
- Two tribes, the **Erenga Kharias** and the **Mankirdias**, inhabit the reserve's forests and practice traditional agricultural activities (the collection of seeds and timber).
- **Vulnerability to Forest Fires:**
 - **Natural:** Natural causes such as lighting or even soaring temperatures can sometimes result in forest fires.
 - **Man-Made Factors:** Instances of poaching and hunting wherein the poachers set a small patch of forest on fire to divert the wild animals can lead to forest fires.

78. (d)

Explanation:

Keoladeo National Park:

- Keoladeo National Park, located in the State of Rajasthan, is an important wintering ground of Palaearctic migratory waterfowl and is renowned for its large congregation of non-migratory resident breeding birds.
- **The Keoladeo National Park is a Ramsar Wetland Site and a World Heritage site.**
- Due to its strategic location in the middle of Central Asian migratory flyway and presence of water, large congregations of ducks, geese, coots, pelicans and waders arrive in the winter. The park is the only known wintering site of the central population of the critically endangered Siberian Crane
- **The wetlands of Keoladeo are not natural**, they are dependent on the monsoon and on water pumped in from outside.
- The wetland is in the shallow depression formed by the confluence of **River Gambhir and River Banganga**, converted into a reservoir by damming the rainwater in monsoons.
- Vegetation is a mosaic of scrub and open grassland that provides habitat for breeding, wintering and staging migratory birds. **Hence, option (d) is correct.**

79. (a)

Explanation:

- The **6th Edition of Shanghai Cooperation Organisation (SCO) Exercise Peaceful Mission: 2021** was hosted by **Russia at Orenburg Region of southwest Russia**.



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- The exercise aims to foster close relations between the SCO Member States and enhance the military leaders' abilities to command multinational military contingents.
- India participated in the 5th edition of the Exercise (2018) for the first time.
- It is a **Multilateral and Joint Counter Terrorism Exercise** conducted biennially as part of military diplomacy between Shanghai Cooperation Organisation (SCO) member states.
- The SCO is an economic and security bloc in which India and Pakistan were admitted as full members in 2017. Its founding members included **China, Russia, Kazakhstan, Kyrgyzstan, Tajikistan, and Uzbekistan.**

80. (c)

Explanation:

- Mangroves are plants that can survive **high salinity, tidal regimes, strong wind velocity, high temperature and muddy anaerobic soil** – a combination of conditions hostile for other plants. **Hence, statement 1 is correct.**
- They are found in the **inter-tidal zones** of sheltered shores, estuaries, creeks, backwaters, lagoons, marshes and mud-flats.
- The mangrove ecosystem constitutes a bridge between terrestrial and marine ecosystems.

The benefits of Mangroves are as follows:

- Mangroves are a **source of firewood, wood products such as timber, poles and posts, and non-wood produce such as fodder, honey, wax, tannin, and dye and plant materials for thatching.**
- Mangrove wetlands and forests act as a **shelter belt against cyclones.**

Mangroves for the Future (MFF):

- It is a unique partner-led initiative to promote investment in coastal ecosystem conservation for sustainable development.
- It is co-chaired by **IUCN**(International Union for Conservation of Nature) and **UNDP** (United Nations Development Programme). **Hence, statement 2 is correct.**
- MFF in India has primary focus on improving the scientific knowledge base to feed the national policies for enhanced management of coastal and marine ecosystems.
- The goal is to promote an integrated ocean-wide approach to coastal management and to building the resilience of ecosystem-dependent coastal communities.

MFF Geographical priority areas:

- The selection is primarily based on the expanse of the mangrove areas and the levels of threats on them.
- The 5 coastal states namely, **West Bengal, Orissa, Gujarat, Andhra Pradesh and Tamil Nadu** have been selected as the geographical priority areas of work under MFF in India.

81. (c)

Explanation:

- The Government of India has instituted a scheme for labeling of **environment friendly products** to be known as **ECO MARK.**
- The scheme is being administered by the **Bureau of Indian Standards.** **Hence, statement 2 is correct.**
- So far, the following product categories have been identified for coverage under this scheme:
 - Soaps and Detergents
 - Paints
 - Paper
 - Plastics
 - Cosmetics
 - Textiles
 - Batteries
 - Wood Substitutes
 - Propellants and Aerosols
 - Food Items (edible oils – including Vanaspati, Tea and Coffee)
 - Electrical and Electronics Goods
 - Packing/Packaging Materials
 - Lubricating/Speciality Oils
 - Drugs
 - Foods Preservatives and Additives
 - Pesticides
 - Leather
- The Scheme is being operated on a **national basis** and provides certification and labeling for **household and other consumer products** which meet certain environmental criteria along with quality requirements prescribed in relevant Indian Standards for the product. **Hence, statement 1 is correct.**
- For the implementation of the scheme, BIS is responsible for the following functions:



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- (i) Assessment of the product for ECO Mark, certification of the product for award of ECO Mark.
- (ii) Renewal, suspension and cancellation of the licence.
- (iii) Products certified as eligible for the ECO Mark shall also carry the ISI Mark **(except for leather)** for quality, safety and performance of the product and shall be licensed to carry the ECO Mark for a prescribed time period after which it shall be reassessed.

82. (b)

Explanation:

- Recently, Meghalaya Enterprise Architecture Project (MeghEA) was launched.
- The MeghEA initiative is spread across **six pillars, i.e. Governance, Human Resources, Entrepreneurship, Primary Sector, Infrastructure, and Environment. Hence, statement 2 is NOT correct.**
- It envisions making Meghalaya a **high-income state by 2030**. The project aims to improve **service delivery and governance for the people** using the power of digital technologies. **Hence, statement 1 is correct.**
- Enterprise Architecture (EA) is how to organise standardised IT (Information Technology) infrastructure to align with business goals.
- The **initiative is in line with the vision of Digital India** that focuses on transforming India into a digitally empowered society and knowledge economy.
- With the notification of India Enterprise Architecture (Indiaorganise standardised) in 2018, the design and implementation of integrated, nationally portable, and scalable digital platforms have gained new momentum.
- During the North East vision document launch in Aug 2018, Meghalaya was selected as the pilot, first pilot State for the preparation of Meghalaya Enterprise Architecture (MeghEA) and design and implementation of MeghEA integrated Finance Solution Architecture.

83. (b)

Explanation:

Arakunomics:

- Arakunomics is a new integrated economic model that **ensures profits for farmers, quality for consumers** through **regenerative agriculture. Hence, option (b) is correct.**

- It is based on work with tribal farmers in Araku for nearly 20 years.
- This economic model is a tribute to the tribal farmers of Araku region for the world class coffee produced and launched in Paris in 2017, as well as for the high carbon landscape transformation they did in over 955 villages, thereby planting 25 million trees.

Naandi Foundation (Arakunomics):

- The organisation was recognised for the application of **Arakunomics model in regions of Araku (Andhra Pradesh), Wardha (Maharashtra) and New Delhi**, leading to the Food Vision 2050 that follows an “**ABC-DEFGH**” framework centring on: **Agriculture, Biology, Compost, Decentralised decision-making, Entrepreneurs, Families, Global Markets, and ‘Headstands (implying innovation)’**.
- Arakunomics success in Araku led to Naandi replicating the model to support the livelihoods of farming communities in the villages of Wardha – infamous for agrarian distress, as well as later in New Delhi, as part of an Urban Farms Co programme.

Regenerative Agriculture

- “**Regenerative Agriculture**” describes farming and grazing practices that, among other benefits, **reverse climate change by rebuilding soil organic matter and restoring degraded soil biodiversity – resulting in both carbon drawdown and improving the water cycle.**
- The key to regenerative agriculture is that it not only “does no harm” to the land but actually improves it, using technologies that regenerate and revitalize the soil and the environment.
- Regenerative agriculture leads to healthy soil, capable of producing high quality, nutrient dense food while simultaneously improving, rather than degrading land, and ultimately leading to productive farms and healthy communities and economies.
- It is dynamic and holistic, incorporating permaculture (sustainable and self-sufficient agricultural ecosystems) and organic farming practices, including conservation tillage, cover crops, crop rotation, composting, mobile animal shelters and pasture cropping, to increase food production, farmers’ income and especially, topsoil.
- It has been promoted to counter loss of the world’s fertile soil and biodiversity, along with the loss of indigenous seeds and knowledge.



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**84. (b)****Explanation:**

- PLAN BEE is an amplifying system imitating the buzz of a swarm of honey bees to **keep wild elephants away from railway tracks. It is a unique strategy adopted by Northeast Frontier Railway (NFR). Hence, option (b) is correct.**
- Speakers that play the sound of bees scare the elephants away. Elephants have long been known to be frightened by the buzzing of bees and stings. Nearly 50 buzzing amplifiers have been deployed as part of “Plan Bee” at a dozen “elephant corridors” in the vast forests of Assam state, home to nearly 6,000 elephants, 20 percent of the country’s total. The buzzing is played as trains approach vulnerable points and can be heard up to half a mile (600 metres) away.

85. (b)**Explanation:****Promotion of Medical Device Parks:**

- The Government of India has taken a key initiative to support the medical devices industry to reach its potential in the coming years identifying the industry as a sunrise sector with great potential for diversification and employment generation. **Hence, statement 1 is correct.**
- Recognising the need for higher levels of investments for the creation of proper infrastructure in the sector, the **Department of Pharmaceuticals** has notified the Scheme for “**Promotion of Medical Device Parks**” with the following objectives:
 - Easy access to standard testing and infrastructure facilities through the creation of **world-class common infrastructure facilities** for increased competitiveness will significantly reduce **the cost of production** of medical devices, leading to better availability and affordability of medical devices in the domestic market.
 - Reaping the benefits arising due to **optimisation of resources** and economies of scale. **Hence, statement 2 is correct.**
- The Medical Devices Parks to be developed under the scheme will provide common infrastructure facilities at one place, thereby creating a robust ecosystem for the medical device manufacturing in the country and reducing the manufacturing cost significantly.

- The total financial outlay of the scheme is Rs. 400 crores, and the tenure of the scheme is from **FY 2020-2021 to FY 2024-2025. Hence, statement 3 is NOT correct.**
- The financial assistance to a selected Medical Device Park would be 70% of the project cost of common infrastructure facilities. In the case of the North-Eastern States and the Hilly States, financial assistance would be 90% of the project cost. Maximum assistance under the scheme for one Medical Device Park would be limited to Rs. 100 crores.
- In total, proposals from 16 States/UTs were received under the scheme. The selection of the States/UTs is based on the challenge method, which is reflected in the evaluation criteria of the scheme. The ranking methodology for States/UTs is based on the parameters prescribed in the scheme guidelines such as **utility charges, State policy incentives, the North-Eastern total area of the park, land lease rate, connectivity of the park, ease of doing business ranking, availability of technical workforce, etc.**
- Based on the evaluation, the proposals of the State Government of Himachal Pradesh, Tamil Nadu, Madhya Pradesh, and Uttar Pradesh have been given “in-principle” approval under the scheme. Further qualitative assessment of the said States in terms of their financial capability, ecosystem attractiveness, and industrial presence also validated the selection of these States.
- The scheme reflects the spirit of **cooperative federalism** where the Central Government and State Governments will partner to develop the Medical Device parks for better sector performance.

86. (b)**Explanation:**

- The term **Marine Protected Areas** include marine reserves, fully protected marine areas, no-take zones, marine sanctuaries, ocean sanctuaries, marine parks, locally managed marine areas, to name a few.
- Many of these have different levels of protection, and the range of activities allowed or prohibited within their boundaries varies considerably too.
- Marine Protected Areas (MPAs) are conserved for a number of reasons including **economic resources, biodiversity conservation, and species protection.**
- They are created by delineating zones with permitted and non-permitted uses within that zone. **Hence, statement 1 is NOT correct.**

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- The IUCN engages in advocating for the expansion of the MPA network through reliable science and by engaging with local stakeholders. **Hence, statement 2 is correct.**
- **Benefits of Marine Protected Areas:**
 - Maintaining biodiversity and providing refuges for endangered and commercial species
 - Protecting critical habitats from damage by destructive fishing practices and other human activities and allowing them to recover
 - Providing areas where fish are able to reproduce, spawn and grow to their adult size
 - Increasing fish catches (both size and quantity) in surrounding fishing grounds
 - Building resilience to protect against damaging external impacts, such as climate change
 - Helping to maintain local cultures, economies, and livelihoods which are intricately linked to the marine environment.

87. (b)**Explanation:****Rajaji Tiger Reserve:**

- Location: **Haridwar (Uttarakhand)**, along the foothills of the **Shivalik range**. It is part of **Rajaji National Park**. **Hence, statement 1 is NOT correct.**
- Background: Rajaji National Park was established in 1983 after amalgamating three sanctuaries in Uttarakhand, i.e. Rajaji, Motichur, and Chila.
 - It was named after the famous freedom fighter C. Rajgopalachari, popularly known as “Rajaji”.
 - It was declared a Tiger Reserve in 2015 as the 48th tiger reserve of the country.
- **Salient Features:**
 - **Flora:** Broadleaved deciduous forests, riverine vegetation, scrubland, grasslands, and pine forests form the range of flora in this park. **Sal (Shorea robusta) is the characteristic dominant tree species. Hence, statement 2 is correct.**
 - **Fauna:** The reserve is home to **more than 50 species**, including tiger, elephant, leopard, Himalayan black bear, sloth bear, jackal, hyena, spotted deer, sambhar, barking deer, nilgai, monkeys, and over 300 species of birds.

- **Rivers: The Ganga and Song rivers** flow through this. **Hence, statement 3 is correct.**

88. (d)**Explanation:**

- **The Forest (Conservation) Act, 1980 (FCA)** governs diversion or use of forest land for non-forest purposes such as industrial or infrastructure projects. A company diverting forest land must provide alternative land for taking up compensatory afforestation. The company should pay for planting new trees in the alternative land provided to the state. The loss of forest ecosystem must also be compensated by paying for net present value of forest (NPV).
- Observing that collected funds for afforestation were underutilised by the states, Supreme Court of India ordered for establishment of Compensatory Afforestation Fund and Compensatory Afforestation Fund Management and Planning Authority (CAMPA) in 2002. **Hence, statement 1 NOT correct.**
- Compensatory afforestation means that every time forest land is diverted for non-forest purposes such as mining or industry, the user agency pays for planting forests over an equal area of non-forest land, or when such land is not available, twice the area of degraded forest land. As per the rules, 90% of the CAF money is to be given to the states while 10% is to be retained by the Centre.
- The funds can be used for treatment of catchment areas, assisted natural generation, forest management, wildlife protection and management, relocation of villages from protected areas, managing human-wildlife conflicts, training and awareness generation, supply of wood saving devices and allied activities. **Hence, statement 2 is NOT correct.**

89. (a)**Explanation:**

- In most ecosystems, all the pyramids, of number, of energy and biomass are upright, i.e., producers are more in number and biomass than the herbivores, and herbivores are more in number and biomass than the carnivores. Also energy at a lower trophic level is always more than at a higher level. **Hence, statement 1 is correct.**

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**There are exceptions to this generalisation.**

- The pyramid of biomass in sea is generally inverted because the biomass of fishes far exceeds that of phytoplankton. **Hence statement 2 is correct.**
- Pyramid of energy is always upright, can never be inverted, because when energy flows from a particular trophic level to the next trophic level, some energy is always lost as heat at each step.
- Each bar in the energy pyramid indicates the amount of energy present at each trophic level in a given time or annually per unit area.

However, there are certain limitations of ecological pyramids such as it does not take into account the same species belonging to two or more trophic levels.

Hence, statement 3 is correct.

- It assumes a simple food chain, something that almost never exists in nature; it does not accommodate a food web.
- Moreover, saprophytes are not given any place in ecological pyramids even though they play a vital role in the ecosystem.

90. (b)**Explanation:**

- On 4 September 2021, Juhu beach, was covered in sticky tarballs that gave off a foul fuel smell.
- **Tarballs:**
 - Tarballs are **dark-coloured, sticky balls of oil** that form when **crude oil** floats on the ocean surface. **Hence, statement 1 is correct.**
 - Tarballs are **formed by weathering of crude oil** in marine environments. **Hence, statement 2 is correct.**
 - They are transported from the open sea to the shores by sea currents and waves, according to a research paper titled, Diversity of bacteria and fungi associated with tarballs: Recent developments and prospects by Laxman Shinde, Varsha & Suneel, V & Shenoy, Belle Damodara (2017), **National Institute of Oceanography (NIO).**
 - Some balls are **as big as a basketball, while others are smaller globules.** Tarballs are usually coin-sized and are found strewn on the beaches. However, they have become as big as basketballs and can **weigh as much as 6-7 kgs over the years.** **Hence, statement 3 is NOT correct.**

- The NIO study states that “**wind and waves tear the oil slick into smaller patches that are scattered over a much wider area. Various physical, chemical and biological processes (weathering) change the appearance of the oil**”.
- It is suspected that the oil comes from the **large cargo ships in the deep sea** and gets pushed to the shore as tarballs during monsoon due to wind speed and direction.

91. (d)**Explanation:**

- **Eutrophication** presents as one of the most serious ecological problems of open water sources such as lakes, oceans and reservoirs.
- It is characterized by **dense algal and plant growth owing to the enrichment by phosphorus and nitrogen nutrients** needed for photosynthesis.
- **Examples** of the plants include algal blooms, Nile cabbage and water hyacinths.
- Eutrophication can also take place naturally over thousands of years as the lakes grow old and get filled with sediments.
- Human activities top the list that speeds up the degree and rate of eutrophication through both point-source and non-point source discharges of the chemical nutrients (phosphates and nitrates) into water systems.

Causes of Eutrophication:

- Discharge of fertilizers (nitrates and phosphates) into water bodies in due to surface run off.
- Concentrated animal feeding operations
- Direct sewage discharge and industrial waste into water bodies
- Aquaculture
- Natural events

Effects of Eutrophication:

1. Threatens the survival of fish and other aquatic life forms
2. Deterioration of water quality and limiting of access to safe drinking water
3. Poisoning and negative impact on human health
4. Depletion of fish species
5. Degradation of recreational opportunities

Hence, statements 1, 2 and 3 are correct.



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**92. (a)****Explanation:****National Board for Wildlife (NBWL):**

- National Board for Wildlife is a statutory Board constituted officially in 2003 under the Wild Life (Protection) Act, 1972. **Hence, statement 1 is correct.**
- The NBWL is chaired by the Prime Minister and is responsible for promotion of conservation and development of wildlife and forests. **Hence, statement 2 is correct.**
- The board is 'advisory' in nature and can only advise the Government on policy making for conservation of wildlife. **Hence, statement 3 is NOT correct.**
- It is an important body because it serves as an apex body for the review of all wildlife-related matters and for the approval of projects in and around national parks and sanctuaries.
- The standing committee of NBWL is chaired by the Minister of Environment Forest and Climate Change.
- The standing committee approves all the projects falling within protected wildlife areas or within 10 km of them.

93. (d)**Explanation:****Web 3.0:**

- Web 3.0 was initially called the **Semantic Web** by World Wide Web inventor **Tim Berners-Lee** and aimed to be a more **autonomous, intelligent, and open internet**. It was established by Ethereum co-founder and former chief technology officer Dr Gavin Wood. **Hence, statement 1 is NOT correct.**
- The Web 3.0 definition can be expanded as follows: data will be interconnected in a decentralised way, in which users will control their own data. **Hence, statement 2 is correct.**
- The next evolution of the Web involves the Semantic Web. The semantic web improves web technologies in demand to create, share and connect content through search and analysis based on the capability to comprehend the meaning of words rather than on keywords or numbers.
- Applying semantics in the Web would enable machines to decode meaning and emotions by analysing data. Consequently, internet users will have a better experience driven by enhanced data connectivity. The new internet

will provide a more personal and customised browsing experience, a brighter and more human-like search assistant, and other decentralised benefits that will help establish a more equitable web. **Hence, statement 3 is correct.**

94. (a)**Explanation:**

- Agasthyamala Biosphere Reserve is located at the southernmost tip of Western ghats.
- Three wildlife sanctuaries, Shendurney, Peppara and Neyyar, are located in the site, as well as the Kalakad Mundanthurai Tiger reserve. **Hence, statement 2 is NOT correct.**
- **Annamalai Tiger Reserve is in Tamil Nadu** where as Agasthyamala in Kerala.
- The reserve has a number of tribal settlements with a total population of 3,000 who rely largely on biological resources for their sustenance. The reserve is home to **Kanikaran tribes** from both Tamil Nadu and Kerala. **Hence, statement 1 is correct.**
- Agasthyamala was included in the list of Man and Biosphere Program of UNESCO in 2016. Khangchendzonga is the latest from India to be included in Man and Biosphere program (2018).

95. (b)**Explanation:**

- **Rhizofiltration** is a **water remediation technique** that involves the **uptake of contaminants by plant roots**. Rhizofiltration is used to reduce contamination in natural wetlands and estuary areas. **Hence, statement 2 is correct.**

Other such techniques are:

- **Mycoremediation** is a form of bioremediation in which fungi are used to decontaminate the area.
- **Mycofiltration** is a similar process to bioremediation, using fungal mycelia to filter toxic waste and micro-organisms from water in soil.
- **Phytoextraction/phytoaccumulation** is the process by which plants accumulate contaminants into the roots and above ground shoots or leaves, thus saving remediation costs.
- **Phytotransformation or phytodegradation** refers to the uptake of organic contaminants from soil, sediments, or water and their transformation to more stable, less toxic, less mobile form.

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○ **Phytostabilization** is a technique in which plants reduce the mobility and migration of contaminated soil. Leachable constituents are adsorbed and bound into the plant structure so that they form unstable mass of plant from which the contaminants will not re-enter the environment.

- Rhizobium is a genus of bacteria whose members are most notable for their ability to establish a symbiotic relationship with leguminous plants, such as peas, soybeans, and alfalfa. It is **NOT** associated with Rhizofiltration. **Hence, statement 1 is NOT correct.**

96. (d)

Explanation:

- The First Movers Coalition is a public-private partnership between the **US State Department, through Special Presidential Envoy for Climate John Kerry, and the World Economic Forum.** It works across sectors and collaborates with leading climate experts and mission-aligned organisations as a platform to accelerate and scale collective impact. **Hence, statement 1 is correct.**
- The First Movers Coalition is a group of forward-thinking companies ready to **jumpstart the global demand for these critical emerging solutions, helping to mobilise the necessary investments and drawdown costs. Members will use their global purchasing power to create new markets for these emerging technologies.** These new demand signals empower suppliers to develop and scale their innovations between now and 2030, helping us reach our global emission targets of reducing emissions to net-zero by 2050. **Hence, statement 2 is correct.**
- The first phase of the First Movers Coalition will focus on “hard-to-abate” sectors, the industries representing more than one-third of the world’s carbon emissions. These include aviation, aluminium, chemicals, concrete, shipping, steel, trucking and direct air capture. **Hence, statement 3 is correct.**

97. (b)

Explanation:

- The red panda is a small arboreal mammal found in the forests of India, Nepal, Bhutan and the northern mountains of Myanmar and southern China. It thrives best at 2,200-4,800m, in mixed deciduous and conifer forests with dense understories of bamboo.

- In India, it is found in Sikkim, western Arunachal Pradesh, Darjeeling district of West Bengal and parts of Meghalaya. It is also the state animal of Sikkim
- Listed as Endangered in the IUCN red list of Threatened Species and under Schedule I of the Indian Wildlife (Protection) Act, 1972, the red panda has the highest legal protection at par with other threatened species.
- **Neora Valley National Park** is one of the richest biological zones in the entire Northeast India and home to elegant **Red Panda** and five species of civet. The park is an important wilderness zone in the Kalimpong hills and also known as birders paradise.
- **Khangchendzonga National Park of Sikkim** is a world heritage site and the first Mixed Heritage site of India. Kanchenjunga Biosphere Reserve is home to amazing list of wild animals like Asiatic Wild Dog, Himalayan Tahr, Snow Leopard, Clouded Leopard and **Red Panda**.

Other National parks where Red panda can be spotted are-

- **Namdapha National Park, Arunachal Pradesh**
 - **Nokrek National Park, Meghalaya**
 - **Singalila National Park, Darjeeling**
- Hence, option (b) is correct.**

98. (c)

Explanation:

- The common components of photochemical smog are ozone, nitric oxide, acrolein, formaldehyde and peroxyacetyl nitrate (PAN). Plants like **Pinus, Juniperus, Quercus, Pyrus and Vitis** can metabolise nitrogen oxide (emitted from combustion of fossil fuels like petrol and diesel). Nitrogen gas is the main gas contributing to the formation of photochemical smog. Peepal does **NOT** metabolise nitrogen oxide. **Hence, statement 1 and 2 are correct.**
- Photochemical smog occurs in warm, dry and sunny climate. The main components of the photochemical smog results from the action of sunlight on unsaturated hydrocarbons and nitrogen oxides produced by automobiles and factories. Photochemical smog has high concentration of oxidising agents and is, therefore, called as oxidising smog.
- Photochemical smog causes serious health problems. Both ozone and PAN act as powerful eye irritants. Ozone and nitric oxide irritate the nose and throat and their high concentration causes headaches, chest pain, dryness of the throat, cough and difficulty in breathing. This smog also causes cracking of rubber and extensive damage to plant life.



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**99. (d)****Explanation:****Ramon Magsaysay:**

- It is named after **Ramon Magsaysay**, the third president of the Philippines. This award is considered Asia's premier prize and highest honour. **Hence statement 1 is correct.**
- The award is internationally-recognized as the Nobel Prize counterpart of Asia and was established in 1957. **Hence, statement 2 is correct.**
- On 31st August every year, it is presented in a formal ceremony in Manila.
- It includes a certificate, a medallion with an embossed image of Ramon Magsaysay, and a cash prize recognising individuals and organisations in Asia, regardless of race, creed, gender, or nationality, who have achieved distinction and have helped others generously without seeking public recognition. **Hence, statement 3 is correct.**
- **Winners of 2021:**
 - Bangladesh's Dr Firdausi Qadri (Affordable Vaccine Champion)
 - Pakistan's Muhammad Amjad Saqib (Poverty Alleviation Visionary)
 - Filipino fisher and community environmentalist Roberto Ballon
 - American Steven Muncy for humanitarian work
 - Refugee assistance and Indonesian torch bearer for investigative journalism, Watchdoc.

100. (c)**Explanation:**

- The total number of Ramsar sites in India is 46 (As of August 2021), the highest in South Asia. **Hence, statement 1 is correct.**

Lonar Lake

- The Lonar lake, situated in the Deccan Plateau's volcanic basalt rock, was created by the impact of a meteor 35,000 to 50,000 years ago.
- The lake is part of Lonar Wildlife Sanctuary which falls under the unified control of the Melghat Tiger Reserve (MTR).
- It is also known as Lonar crater and is a notified National Geo-heritage Monument. Geo-heritage refers to the geological features which are inherently or culturally significant offering insight to earth's

evolution or history to earth science or that can be utilized for education.

- It is the second Ramsar site in Maharashtra after Nandur Madhmeshwar Bird Sanctuary in Nashik district.
- The water in the lake is highly saline and alkaline, containing special microorganisms like anaerobes, Cyanobacteria and phytoplankton.

Kabartal Wetland:

- Also known as Kanwar Jheel, it covers 2,620 hectares of the Indo-Gangetic plains in the Begusarai district of Bihar.
- It acts as a vital flood buffer for the region besides providing livelihood opportunities to local communities.
- Significant biodiversity is present, with 165 plant species and 394 animal species recorded, including 221 bird species. It is also a valuable site for fish biodiversity with over 50 species documented.
- **It is an important stopover along the Central Asian Flyway, with 58 migratory waterbirds using it to rest and refuel. Hence, statement 3 is correct.**
- Five critically endangered species inhabit the site, including three vultures – the **red-headed vulture** (*Sarcogyps calvus*), **white-rumped vulture** (*Gyps bengalensis*) and **Indian vulture** (*Gyps indicus*) – and two waterbirds, the **sociable lapwing** (*Vanellus gregarius*) and **Baer's pochard** (*Aythya baeri*).
- Major threats to the Site include water management activities such as drainage, water abstraction, damming and canalization.

Asan Conservation Reserve (ACR):

- ACR is a 444-hectare stretch of the Asan River running down to its confluence with the **Yamuna River in Dehradun district of Uttarakhand. It is Uttarakhand's first Ramsar Site. Hence, statement 2 is NOT correct.**
 - The damming of the River by the Asan Barrage in 1967 resulted in siltation above the dam wall, which helped to create some of the Site's bird-friendly habitats.
- These habitats support 330 species of birds including the critically endangered **red-headed vulture** (*Sarcogyps calvus*), **white-rumped vulture** (*Gyps bengalensis*) and **Baer's pochard** (*Aythya baeri*).
- Other non-avian species present include 49 fish species, one of these being the endangered Putitora mahseer (*Tor putitora*). Fish use the site for feeding, migration and spawning.

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